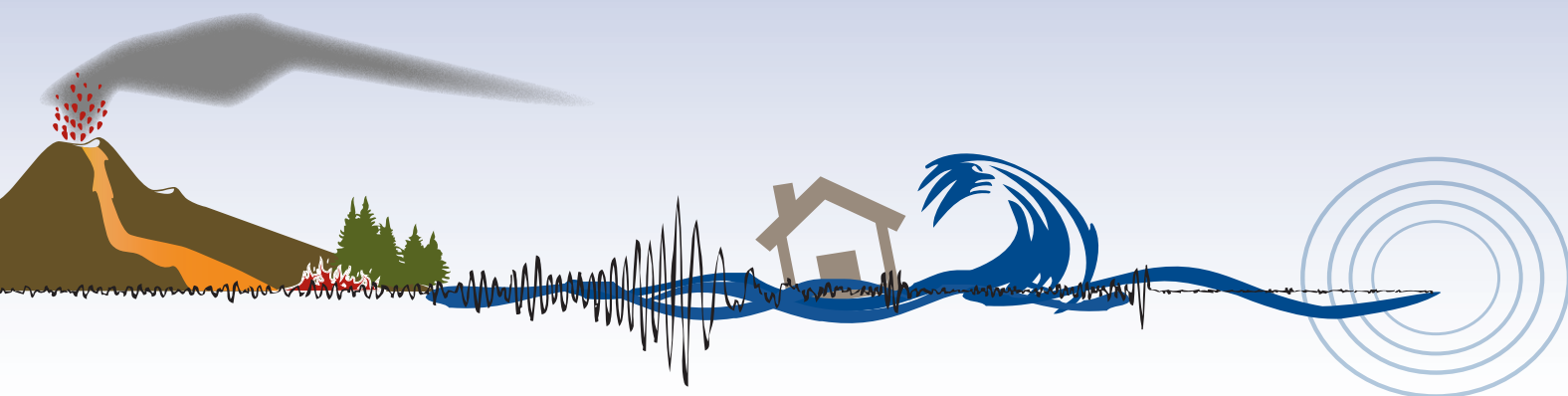


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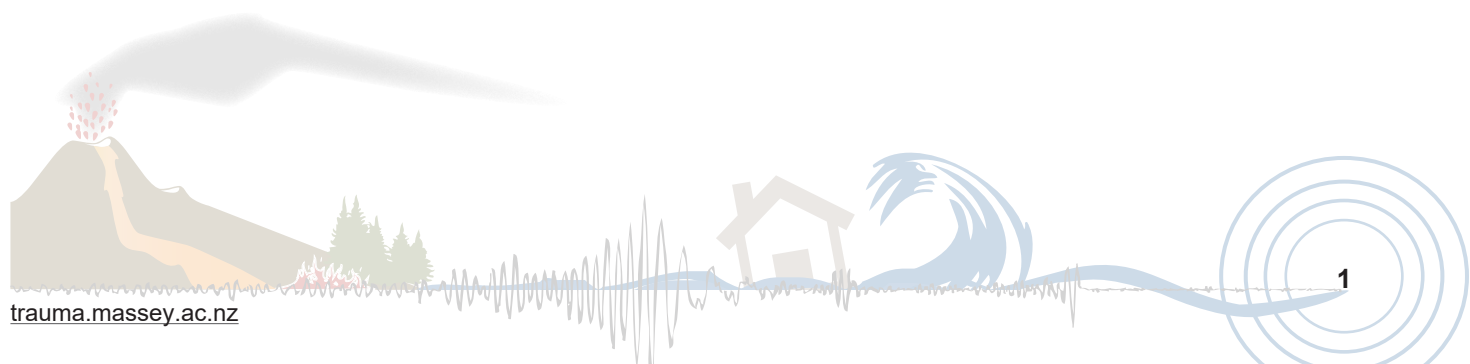
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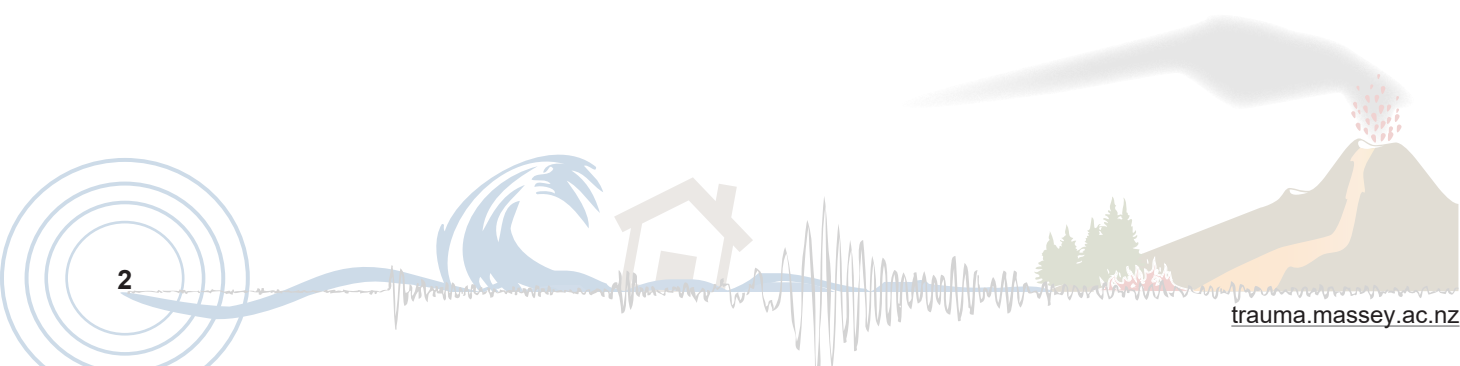
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Factors distinguishing treatment-seeking individuals and individuals who self-identified as coping well following the Canterbury Earthquake Sequence

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Abstract

Following potentially traumatic events, most individuals do not develop mental health issues and some even experience high levels of psychological functioning. Others, however, develop mental health conditions such as posttraumatic stress disorder (PTSD), depression, or anxiety disorders, and require specialist mental health treatment. Few studies have directly compared individuals at these opposite ends of the trauma response spectrum following the same event. The current study identified factors distinguishing individuals seeking treatment for severe earthquake-related distress (n = 184) and those who did not develop earthquake-related distress and self-identified as coping well (n = 101) following the 2010/11 Canterbury earthquake sequence. Participants completed measures of sociodemographic

factors, mental health history, earthquake exposure, life events, and current psychological functioning. In univariate analyses, the groups reported comparable levels of objective earthquake exposure, but treatment-seeking participants reported significantly more pre-earthquake mental disorders, peritraumatic distress (i.e., emotional and physiological distress experienced during the earthquake), exposure-related distress (i.e., distress associated with earthquake-related traumatic events), higher number of life events (e.g., marriage, death of a family member) experienced in the past 5 years, and greater perceived difficulty associated with life events. Treatment-seeking participants were also significantly younger, more likely to be female, less likely to have a tertiary educational qualification, and less likely to be in a relationship. In multivariate analyses, peritraumatic distress and education level were the only factors that uniquely distinguished the groups. Findings emphasise the importance of peritraumatic experiences and education level in the development of distinct post-trauma outcomes among individuals exposed to similar degrees of disaster exposure. Subjective peritraumatic responses are likely to be useful to consider in screening efforts following disasters.

Keywords: Posttraumatic stress disorder, post-trauma outcomes, disaster, earthquake

Over 70% of people are exposed to potentially traumatic events during their lifetime (Benjet et al., 2015). Trauma exposure increases risk for mental disorders, including posttraumatic stress disorder (PTSD), mood, anxiety, and substance use disorders (Auxéméry, 2018). There has been a focus on PTSD following potentially traumatic events, which is characterised by intrusion, avoidance, and hyperarousal symptoms, alongside disturbed cognition, mood, and functioning (Asmundson et al., 2000). Despite the high prevalence of trauma exposure, relatively few individuals develop PTSD (lifetime prevalence is 8.3%; Kilpatrick et al., 2013) and most people experience only mild trauma responses, with few or no PTSD symptoms (Bonanno, 2004).

A range of pre-, peri-, and post-trauma factors have been found to influence risk for the development of PTSD. Pre-trauma factors are pre-existing characteristics unrelated to the traumatic event, including sociodemographic factors and factors relating to pre-trauma mental health and trauma exposure. Female gender is a robust risk factor for the development of PTSD (Tortella-Feliu et al., 2019). Women are more likely than men to experience events conferring greater risk for PTSD (e.g., sexual assault) and are more likely to experience other risk factors, such as heightened peritraumatic distress (i.e., emotional and physiological distress experienced during the traumatic event; Christiansen & Hansen, 2015). Other sociodemographic factors, including younger age and lower education, were identified as risk factors in earlier meta-analyses (Brewin et al., 2000), but these findings have not been replicated in larger, more recent syntheses (Tortella-Feliu et al., 2019). Higher rates of PTSD have also been observed among ethnic minority groups (Asnaani & Hall-Clark, 2017). The presence of pre-trauma mental disorder, particularly prior PTSD, is a well-established risk factor for PTSD after subsequent exposure (Brewin et al., 2000; Fernandez et al., 2016). Prior trauma exposure has also been found to increase risk for PTSD (Ozer et al., 2003).

Peritraumatic factors relate to an individual's experiences during and immediately after the traumatic event, including the severity of objective trauma exposure and subjective responses to the event. Greater degree of objective exposure is a risk factor for PTSD (Brewin et al., 2000). In the context of a disaster, for example, people who experienced bereavement, sustained injuries, or lost possessions are considered to have a higher degree of objective exposure (Elal & Slade, 2005). The importance of subjective experiences has been noted in addition to objective experiences. People who experience more intense subjective emotional and physiological distress during a traumatic event (i.e., greater peritraumatic distress, panic, and dissociation) are more likely to develop PTSD (Vance et al., 2018). It is hypothesised this may be due to the way these responses disrupt the encoding of the event into memory and contribute to negative post-trauma cognitions (Ehlers & Clark, 2000).

Post-trauma factors are those that occur after the traumatic event, including degree of post-trauma social support, life stress, and resource loss. Reduced social support has been identified as a risk factor for PTSD in earlier meta-analyses (Ozer et al., 2003); however, this finding has not been replicated in more recent syntheses (Tortella-Feliu et al., 2019). Increased life stress following

trauma exposure is a risk factor for PTSD (Brewin et al., 2000), perhaps because stress depletes the psychosocial resources people utilise to cope with traumatic events. Relatedly, conservation of resources theory (Hobfoll et al., 2006) posits that people who lose a greater degree of resources following a traumatic event (e.g., personal possessions, psychological coping resources, valued social roles) are more likely to develop PTSD.

Disasters triggered by natural processes are collectively experienced and often expose entire communities to a series of potentially traumatic events. Individuals exposed to the same disaster may have similar objective experiences but can experience markedly different outcomes. Some individuals do not develop disaster-related mental health issues and, additionally, report high levels of psychological functioning despite substantial disaster exposure (Bonanno et al., 2006). Others, however, will require specialist mental health treatment for severe and ongoing disaster-related distress (North & Pfefferbaum, 2013). While multiple studies have examined factors associated with the development of PTSD symptoms among people exposed to the same disaster, few studies have directly compared individuals at opposite ends of the trauma response spectrum: those seeking treatment for disaster-related distress and those who exhibit high levels of functioning and the absence of mental health problems, despite substantial levels of exposure severity. Factors distinguishing these groups could be used to identify individuals most at risk of requiring specialist mental health services, which has potential to inform outreach and service planning efforts following disasters.

In 2010 and 2011, a series of major earthquakes occurred in the Canterbury region in Aotearoa New Zealand, affecting a large proportion of the community and resulting in 185 deaths, thousands of injuries, and widespread damage (Ardagh et al., 2012). Many Canterbury residents coped well, but others sought treatment for severe, ongoing, earthquake-related distress. The current study examined factors distinguishing people who sought treatment compared to a control group who did not develop mental health issues and coped well following the Canterbury earthquakes, including factors relating to sociodemographics, pre-earthquake mental health, degree of trauma exposure, peritraumatic distress, and life events.

Method

Study Design

The study utilised an observational cross-sectional design, in which earthquake-exposed participants with distinct post-earthquake outcomes were retrospectively assessed at one time point following the earthquakes.

Participants

Earthquake-Exposed Treatment-Seeking Group

This group comprised 184 Canterbury residents referred to a specialist mental health service that was initiated for the treatment of severe, ongoing earthquake-related distress (i.e., PTSD or adjustment disorder with earthquake-related anxiety). The service was the main provider of publicly-funded mental health treatment for those with severe levels of distress related to the earthquakes. Participants were referred by general medical practitioners, other mental health services, or psychiatrists and psychologists working in private practice. People were invited to take part in the study during initial face-to-face assessments (see Procedure section), but participation was not required to receive treatment. Inclusion criteria were age 18 years and above and presence of earthquake-related DSM-IV PTSD or adjustment disorder with earthquake-related anxiety. Those for whom the primary earthquake-related disorder was not PTSD or adjustment disorder (e.g., depression) were not treated in the service and were referred to existing tertiary mental health services. Other exclusion criteria were current mania, psychotic depression, severe alcohol or drug disorders/problems, and active suicidality. Treatment-seeking participants were recruited over a period of 3 years and 8 months, between August 2011 and April 2015, approximately 1 to 4.5 years after the onset of the earthquake sequence. The flow of treatment-seeking participants through the recruitment process is displayed in Appendix A. Of the 253 people who were referred to the service and attended initial clinical assessments, 184 (72.7%) were suitable for the service and consented to participate in the study.

Earthquake-Exposed Control Group

This group comprised 101 Canterbury residents who did not develop earthquake-related mental health issues and who self-identified (at the time of recruitment) as having coped well following the earthquakes, despite moderate to high exposure to earthquake-related events such as sustaining physical injury, seeing buildings falling down, seeing injured people and dead bodies, or damage to

property and possessions (henceforth referred to as the “control group” or “control participants”). Control participants, recruited from October 2012 to November 2013, responded to publicity in newspapers, flyers posted in public places, community websites, and word of mouth. Potential participants were invited to respond by contacting the research team directly, either by telephone or email. Control participants were excluded if they had developed an earthquake-related mental disorder or received treatment for earthquake-related distress. Participants were not excluded if they had current mental disorders that were unrelated to and pre-dated the earthquake sequence. The research team met regularly to discuss and determine the eligibility of control participants, after both the initial screening and interview assessments. Those who did not experience substantial earthquake exposure or who were assessed as not coping well were excluded. The flow of control participants through the recruitment process is displayed in Appendix B.

Procedure

Both control and treatment-seeking participants were screened by telephone for eligibility before attending face-to-face assessments, during which participants were provided with verbal information about the study alongside a written information sheet. Once participants had received information about the study, informed written consent was obtained prior to data collection. The face-to-face assessment consisted of the Mini International Neuropsychiatric Interview (MINI; Sheehan et al., 1998) to assess the presence of mental disorders. Participants also completed pen-and-paper demographic and self-report measures (described below). Participants were allowed to skip questions they did not want to answer. Treatment-seeking participants were assessed by experienced mental health clinicians (e.g., clinical psychologists, experienced mental health nurses); control participants were assessed by research assistants or post-graduate students trained to administer the MINI. Participants did not receive incentives for participation.

Measures

Sociodemographic Information. Participants provided sociodemographic information relating to gender, age, relationship status, education, and ethnicity.

Mini-International Neuropsychiatric Interview (MINI). The MINI is a structured diagnostic interview used to assess the presence of common mental disorders including PTSD, adjustment disorder, mood, anxiety,

substance use, and eating disorders (Sheehan et al., 1998). Current and pre-earthquake diagnoses were assessed. Pre-earthquake diagnoses were based on post-earthquake recall.

Posttraumatic Stress Disorder Checklist—Specific (PCL-S). The PCL-S is a 17-item scale assessing DSM-IV PTSD symptoms relating to a specific event (in this case, the Canterbury earthquakes; Blanchard et al., 1996). Five items assess intrusion symptoms ($\alpha = .95$; this value, and subsequent α values, are from the current sample), seven items assess avoidance symptoms ($\alpha = .91$), and five items assess hyperarousal symptoms ($\alpha = .93$). Participants rate items from 1 = *not at all* to 5 = *extremely*. A total severity score was calculated by summing all item scores ($\alpha = .97$; possible range = 17–85; this and all other ranges reported in this section reflect possible ranges for the scale). Higher scores reflect more severe PTSD symptoms.

Connor-Davidson Resilience Scale (CD-RISC). The CD-RISC (Connor & Davidson, 2003) is a 25-item scale assessing components of resilience such as adaptability to change, engaging the support of others, and self-efficacy. Participants rate items from 0 = *not true at all* to 4 = *true nearly all the time*. Item scores are summed to produce a total score ($\alpha = .95$; range = 0–100). Higher scores reflect greater resilience.

Depression Anxiety Stress Scales (DASS-21). The 21-item DASS-21 measures past-week depression, anxiety, and stress (Lovibond & Lovibond, 1995). Seven items pertain to each of the depression ($\alpha = .95$), anxiety ($\alpha = .93$), and stress ($\alpha = .95$) subscales. Items are rated from 0 = *did not apply to me at all* to 3 = *applied to me very much, or most of the time*. Subscale items are summed and multiplied by two to produce subscale scores (range = 0–42). Higher scores indicate greater depression, anxiety, and stress.

Social Adjustment Scale (SAS). The SAS is a 45-item scale measuring social functioning over the past 2 weeks (Weissman & Bothwell, 1976) across seven domains: work outside the home (six items; $\alpha = .92$), household tasks (six items; $\alpha = .83$), social and leisure activities (nine items; $\alpha = .83$), extended family (seven items; $\alpha = .64$), partner (ten items; $\alpha = .94$), parental (four items; $\alpha = .94$), and family unit (three items; $\alpha = .88$). Items are rated from 1 to 5. A total score reflecting social adjustment across all domains was produced by averaging all item scores ($\alpha = .91$; range = 1–5). Higher scores indicate poorer social adjustment.

Buss-Perry Aggression Questionnaire (BPAQ).

The current study used a shortened 12-item version (Bryant & Smith, 2001) of the BPAQ (Buss & Perry, 1992) to measure aggression. The measure contains four subscales, each containing three items: physical aggression ($\alpha = .70$), verbal aggression ($\alpha = .89$), anger ($\alpha = .86$), and hostility ($\alpha = .91$). Items are rated from 1 = *extremely uncharacteristic of me* to 5 = *extremely characteristic of me* and summed to produce a total score ($\alpha = .91$; range = 12–60). Higher scores indicate greater aggression.

Traumatic Exposure Severity Scale (TESS). Degree of earthquake exposure was assessed using a modified version of the TESS (Elal & Slade, 2005). The original TESS is a 24-item measure assessing resource loss, damage to home and goods, personal harm, concern for significant others, and exposure to the grotesque. Items were removed, modified, and added to reflect the experiences of Canterbury residents. The modified version deleted one item (“Did you experience the odour of dead bodies in the days following the earthquake?”) and split one item (“Were you buried under rubble/trapped for a period of time”) into two items (“Were you buried under rubble for a period of time after the earthquake?”, “Were you trapped in a building for a period of time after the earthquake?”). Six items were added relating to acute exposure (e.g., “Did you see buildings falling down as a result of the earthquake?”) and nine items were added relating to secondary stressors (e.g., “Have you lost your job since the earthquake?”), creating a 39-item measure. For each item, participants indicate whether they experienced the event, and if so, how distressing the event was (1 = *not at all* to 5 = *extremely*). An occurrence score was produced by summing the number of events experienced. A distress score was produced by summing distress scores. Higher occurrence and distress scores reflect greater earthquake exposure and exposure-related distress, respectively.

Peritraumatic Distress Inventory (PDI). The PDI is a 13-item scale measuring distress during and immediately after a traumatic event (Brunet et al., 2001). The measure includes items pertaining to both emotional distress (e.g., “I felt helpless”; “I felt afraid for my own safety”) and physiological distress (e.g., “I had physical reactions like sweating, shaking, and my heart pounding”; “I felt I might pass out”). Participants identified the earthquake they found most distressing as the index event. Items are rated from 0 = *not at all* to 4 = *extremely true*. The total score is the mean item score ($\alpha = .92$; range = 0–4). Higher scores reflect greater peritraumatic distress.

Life Events Scale (LES). An adaptation of the Crisis in Family Systems Revised Questionnaire (Shalowitz et al., 1998) was used to measure number of life events experienced in the past 5 years. The measure includes 64 items relating to 11 domains (e.g., financial, legal, career). Two items were added in the adapted version ("Did a family pet die?" and "Did a close family member divorce or separate?"). Total number of life events was calculated by summing each event experienced. The perceived difficulty of experienced events was rated from 1 = *not at all* to 4 = *a lot*. A total difficulty score was the mean of the difficulty ratings. For treatment-seeking participants recruited at the onset of the mental health service, the 5-year period mostly covered events occurring before the earthquakes. For those recruited towards the end of the service, the 5-year period mostly covered events occurring after the earthquakes.

Ethical Approval

The study was approved by the National Health and Disability Ethics Committee (URA/11/EXP/027, URA/12/03/011).

Data Analysis

Descriptive Analyses

All statistical analyses were conducted using SPSS v. 28.01. Demographic and MINI data were summarised using standard descriptive statistics including means and standard deviations for continuous measures and frequencies and percentages for categorical measures.

Group Differences in Current Psychological Functioning

Univariate logistic regression analyses were used to examine group differences on measures of current psychological functioning. These analyses were conducted to ensure that the treatment-seeking and coping well groups were indeed different in terms of post-earthquake psychological functioning (as intended), and to provide a descriptive overview of the groups' current psychological functioning. Posttraumatic stress symptoms (PCL), resilience (CD-RISC), depression, anxiety, and stress symptoms (DASS), social functioning (SAS), and aggression (BPAQ) were individually entered as independent variables, with group membership (treatment-seeking vs. control) as the dependent variable.

Main Analyses

A series of univariate logistic regression analyses was first used to identify factors distinguishing treatment-seeking and control participants. Number of pre-earthquake

mental disorders (MINI), objective earthquake exposure severity and exposure-related distress (TESS), peritraumatic distress (PDI), number of life events experienced in the past 5 years and difficulty associated with life events (LES), and sociodemographic factors (gender, age, education level, and relationship status) were each individually tested for association with post-trauma outcome (treatment-seeking vs. control). Univariate analyses were conducted prior to multivariate analyses to understand the association between individual factors and post-trauma outcome, and to identify candidate variables for multivariate analyses.

Factors identified as potentially associated with post-trauma outcome (using a significance threshold of $p < .10$) were considered candidate predictors. A higher significance threshold (i.e., $p < .10$ rather than the conventional $p < .05$) was used when identifying candidate predictors, as the primary aim of univariate analyses was to identify potential relationships, rather than to draw inferences about factors independently distinguishing the treatment-seeking and control groups. Multivariate logistic regression models were then constructed to estimate associations between candidate predictors and post-trauma outcome (treatment-seeking vs. control). To identify unique predictors, candidate predictors were entered into two separate regression models using forwards and backwards stepwise approaches. Predictors significantly associated with post-trauma outcome ($p < .05$) in both models were then entered simultaneously into a final logistic regression model. To account for differences in participants' timing of assessment, the final model was re-run adjusting for time elapsed between the onset of the earthquakes (04 September 2010) and date of assessment (less than 2.5 years vs. more than 2.5 years). To assess performance of the final model, predicted probabilities were generated from this model and used to calculate the area under the receiver operating curve (AUROC) predicting post-trauma outcome.

Missing Data

In the treatment-seeking group, missing data rates for variables ranged from 0% to 38.6%. These participants came from a clinical service operating in a post-earthquake environment and it was not always possible to collect full data for each participant. In the control group, missing data rates for variables ranged from 0% to 15.8%. In univariate logistic regression analyses, all cases with valid values on the predictor of interest were included. In multivariate analyses, a listwise deletion approach was implemented whereby any cases with missing values were excluded from analysis. Sample sizes for the individual measures are shown in tables.

Results

Descriptive Analyses

In the treatment-seeking group ($n = 184$), 153 participants (83.2%) were female, 30 (16.3%) were male, and one (0.5%) identified as a transgender woman. The mean age was 43.7 years ($SD = 13.27$). Ninety-six participants were in a relationship (52.5%; one did not have available data for relationship status) and 55 had a tertiary educational qualification (36.2%; 32 did not have available data for education level). Most treatment-seeking participants reported New Zealand European ethnicity ($n = 161$, 87.5%). In the control group ($n = 101$), 67 participants (66.3%) were female, and 34 (33.7%) were male. The mean age was 50.1 years ($SD = 11.02$). Seventy participants were in a relationship (69.3%) and 77 had a tertiary educational qualification (76.2%). Most participants reported New Zealand European ethnicity ($n = 80$, 79.2%).

Table 1 displays current and pre-earthquake mental disorders from the MINI. Most treatment-seeking participants had earthquake-related PTSD (79.0%). The remainder (21.0%) did not meet full criteria for PTSD and were diagnosed with adjustment disorder with earthquake-related anxiety. Most treatment-seeking participants had at least one pre-earthquake mental disorder (79.9% of the 174 participants with available pre-earthquake MINI data), with mood disorders (59.1%), generalised anxiety disorder (31.5%), and panic disorder (29.9%) the most common. Rates of current co-occurring disorders were high in this group, with mood disorders (74.0%), generalised anxiety disorder (48.6%), and panic disorder (45.9%) the most common. Low rates of current disorders were found in the control group. Six participants had a current disorder (four with agoraphobia, one with panic disorder, and one with a substance use disorder) that pre-dated the earthquakes. Thirty-seven percent had a pre-earthquake mood disorder and 33.0% had a pre-earthquake substance use disorder. Rates of other pre-earthquake disorders ranged from 1.1% for adjustment disorder and generalised anxiety disorder to 14.6% for PTSD.

Group Differences in Current Psychological Functioning

Table 2 displays summary statistics and results from univariate logistic regression analyses for variables related to current psychological functioning. Compared to control participants, treatment-seeking participants reported greater PTSD symptom severity (PCL-S), less resilience (CD-RISC), greater depression, anxiety, and stress symptoms (DASS), poorer social adjustment (SAS), and greater aggression (BPAQ). All current psychological functioning factors were significantly associated with group membership (all $p < .001$).

Table 1

Current and Pre-Earthquake Disorders from the MINI for the Treatment-Seeking and Control Groups

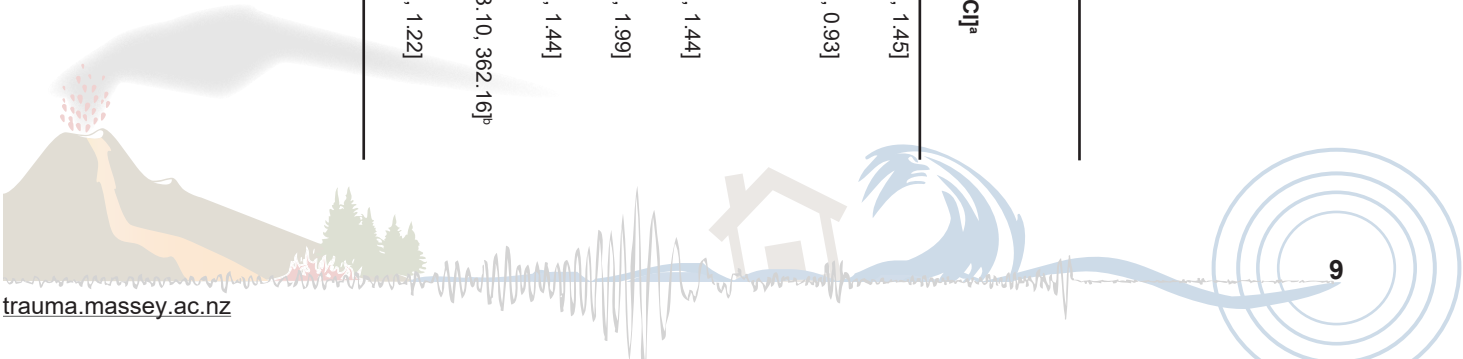
Disorder	Treatment-seeking group ^a		Control group ^b	
	<i>n</i>	%	<i>n</i>	%
PTSD				
Current	143	79.0	0	0.0
Pre-earthquake	46	26.1	13	14.6
Adjustment disorder				
Current	39	21.5	0	0.0
Pre-earthquake	4	2.3	1	1.1
Panic disorder				
Current	83	45.9	1	1.0
Pre-earthquake	53	29.9	5	5.7
Agoraphobia				
Current	48	26.5	4	4.0
Pre-earthquake	29	16.3	5	5.7
Social phobia				
Current	49	27.1	0	0.0
Pre-earthquake	46	25.8	5	5.7
Obsessive compulsive disorder				
Current	16	8.8	0	0.0
Pre-earthquake	16	9.0	3	3.4
Generalised anxiety disorder				
Current	88	48.6	0	0.0
Pre-earthquake	56	31.5	1	1.1
Mood disorders^c				
Current	134	74.0	0	0.0
Pre-earthquake	104	59.1	32	36.8
Substance use disorders				
Current	18	9.9	1	1.0
Pre-earthquake	24	13.4	29	33.0

Note. Low levels of current disorders in the control group were due to exclusion criteria for control participants. ^a *n* ranged from 176 to 181, ^b *n* ranged from 87 to 101, ^c Major depression or dysthymia

Table 2
Descriptive Statistics and Results of Univariate Logistic Regression Analyses Examining Group Differences in Current Psychological Functioning

	Treatment-seeking group		Control group		B	SE	p	OR [95% CI] ^a
	Mean	SD	Mean	SD				
Posttraumatic Stress Disorder Checklist (<i>n</i> = 219)	54.70	14.93	21.66	4.89	0.28	0.05	< .001	1.33 [1.21, 1.45]
Connor-Davidson Resilience Scale (<i>n</i> = 216)	52.81	18.09	75.83	12.03	-0.10	0.01	< .001	0.91 [0.88, 0.93]
Depression Anxiety Stress Scales								
Depression subscale score (<i>n</i> = 284)	21.34	11.92	2.67	3.69	0.29	0.04	< .001	1.33 [1.24, 1.44]
Anxiety subscale score (<i>n</i> = 284)	21.45	10.54	1.19	2.42	0.54	0.08	< .001	1.71 [1.47, 1.99]
Stress subscale score (<i>n</i> = 284)	26.33	9.66	5.15	5.09	0.29	0.04	< .001	1.34 [1.25, 1.44]
Social Adjustment Scale (<i>n</i> = 217)	2.43	0.51	1.63	0.29	4.70	0.61	< .001	109.48 [33.10, 362.16] ^b
Buss-Perry Aggression Questionnaire (<i>n</i> = 219)	23.44	9.43	15.58	5.51	0.15	0.03	< .001	1.16 [1.10, 1.22]

Note. ^a ORs relate to a 1-unit increase on each scale. ^b OR is large due to the small range associated with this variable



Main Analyses

Univariate Logistic Regression Analyses

Table 3 displays summary statistics and results from univariate logistic regression analyses comparing the groups on gender, age, education level, relationship status, number of pre-earthquake mental disorders (MINI), objective earthquake exposure severity and exposure-related distress (TESS), peritraumatic distress (PDI), number of life events experienced in the past 5 years, and difficulty associated with life events (LES). The treatment-seeking group was significantly younger than the control group ($p < .001$) and was characterised by a greater proportion of female participants ($p = .001$), and a smaller proportion of participants with a tertiary educational qualification ($p < .001$) and participants in a relationship ($p = .006$). Treatment-seeking and control participants reported comparable degrees of objective earthquake exposure ($p = .058$). Treatment-seeking participants had significantly more pre-earthquake disorders, reported greater exposure-related distress (TESS), greater peritraumatic distress (PDI), and greater exposure to and difficulty associated with life events (LES) compared to control participants (LES; all $p \leq .001$).

Multivariate Logistic Regression Analyses

Using a significance threshold of $p < .10$, the following factors were entered as independent variables into multivariate logistic regression analyses to identify factors uniquely associated with group membership: age, gender, education level, relationship status, number of pre-earthquake disorders, objective earthquake exposure severity and exposure-related distress, peritraumatic distress, number of life events in past 5 years, and difficulty associated with life events. Both forwards and backwards models converged to the same significant ($p < .05$) factors, education level (proportion of participants with a tertiary educational qualification) and peritraumatic distress (as measured by the PDI). The final model including education level and peritraumatic distress (with 116 treatment-seeking participants and 101 control participants) was statistically significant ($\chi^2(2) = 139.02, p < .001$). Having a tertiary educational qualification was associated with decreased odds of membership in the treatment-seeking group (OR = 0.26, 95% CI = 0.12, 0.60). Greater peritraumatic distress was associated with membership in the treatment-seeking group. For every 1-point increase in PDI score, the odds of treatment-seeking increased approximately 8.0 times (OR = 8.08, 95% CI = 4.68, 13.94). This final

model including education level and the PDI total score demonstrated excellent predictive accuracy (AUROC = .91, $p < .001$, 95% CI = .87, .95). Adjusting the final model for time elapsed between the onset of the earthquakes and assessment (less than 2.5 years vs. more than 2.5 years) did not substantially alter the results of the model.

Some treatment-seeking participants did not have available data for education level ($n = 32$) or the PDI ($n = 68$). Treatment-seeking participants with and without available education and PDI data were compared on several factors for which complete, or almost complete, data were available, including some sociodemographic (gender, age, ethnicity) and clinical (DASS depression, anxiety, and stress subscales, number of current disorders) characteristics. Treatment-seeking participants with and without available education and PDI data did not differ significantly on any of these variables (all $p \geq .259$), suggesting treatment-seeking participants with available education and PDI data were comparable to those without available data for these variables.

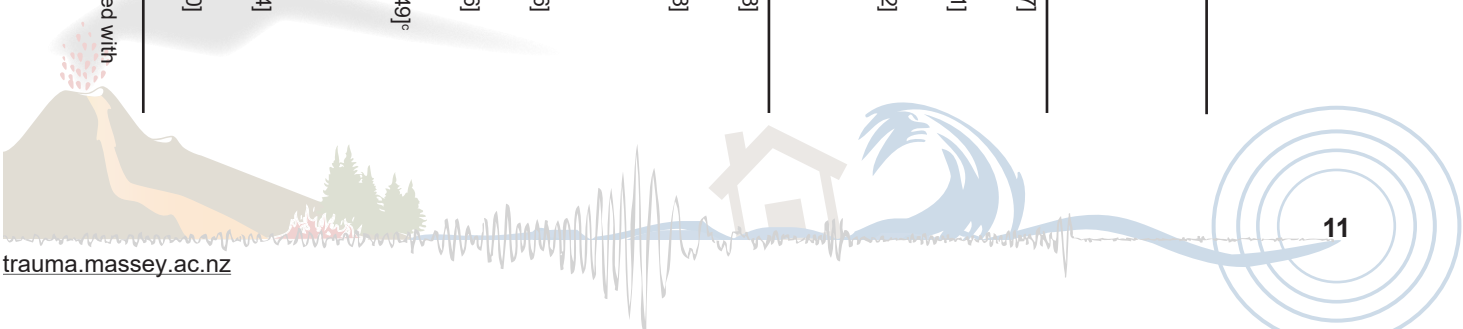
Discussion

The current study identified factors distinguishing individuals who experienced distinct outcomes following exposure to the Canterbury earthquakes: those who sought treatment for severe earthquake-related distress, and those who did not develop earthquake-related mental health issues and self-identified as coping well. Univariate analyses indicated the groups experienced comparable degrees of objective earthquake exposure, but treatment-seeking participants reported more pre-earthquake mental disorders, greater exposure-related distress and peritraumatic distress, more life events experienced in the past 5 years, and greater difficulty associated with life events. Treatment-seeking participants were also significantly younger on average, were more likely to be female, less likely to have a tertiary educational qualification, and less likely to be in a relationship. In multivariate analyses, only peritraumatic distress and education level were uniquely associated with group membership, suggesting the remaining factors were not associated with post-trauma outcome beyond their association with peritraumatic distress and education level. These two variables demonstrated excellent predictive accuracy, indicating peritraumatic distress and education level were important factors distinguishing treatment-seeking and control participants.

Table 3
Descriptive Statistics and Results of Main Univariate Logistic Regression Analyses

	Treatment-seeking group		Control group		B	SE	p	OR [95% CI] ^b
	n	%	n	%				
Gender (female; <i>n</i> = 284) ^a	153	83.6	67	66.3	0.95	0.29	.001	2.59 [1.47, 4.57]
Education (with tertiary qualification; <i>n</i> = 253)	55	36.2	77	76.2	-1.73	0.29	<.001	0.18 [0.10, 0.31]
Relationship status (in a relationship; <i>n</i> = 284)	96	52.5	70	69.3	-0.72	0.26	.006	0.49 [0.29, 0.82]
	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>	<i>B</i>	<i>SE</i>	<i>p</i>	<i>OR [95% CI]^b</i>
Age (years; <i>n</i> = 285)	43.74	13.27	50.13	11.02	-0.04	0.01	<.001	0.96 [0.94, 0.98]
Number of pre-earthquake disorders (<i>n</i> = 260)	2.21	1.92	1.28	1.50	0.34	0.09	<.001	1.40 [1.17, 1.68]
Traumatic Exposure Severity Scale								
Occurrence score (<i>n</i> = 214)	12.56	3.83	11.58	3.55	0.07	0.04	.058	1.07 [1.00, 1.16]
Distress score (<i>n</i> = 214)	49.54	22.78	33.79	17.34	0.04	0.01	<.001	1.04 [1.02, 1.06]
Peritraumatic Distress Inventory (<i>n</i> = 217)	2.48	0.86	1.04	0.63	2.19	0.28	<.001	8.95 [5.17, 15.49] ^c
Life Events Scale								
Number of life events in past 5 years (<i>n</i> = 215)	11.61	6.73	8.84	5.10	0.08	0.03	.001	1.08 [1.03, 1.14]
Difficulty score (<i>n</i> = 215)	2.65	1.11	2.08	0.67	0.82	0.19	<.001	2.26 [1.55, 3.30]

Note. ^a Analysis excludes one participant who identified as a transgender woman. ^b ORs relate to a 1-unit increase on each scale. ^c ORs is large due to the small range associated with this variable.



The finding that emotional and physiological distress experienced during the earthquakes (i.e., peritraumatic distress) emerged as an important distinguishing factor is consistent with theoretical models of PTSD highlighting the role of subjective experiences in the development of the disorder. Ehlers and Clark's (2000) cognitive model of PTSD posits that PTSD develops when a traumatic event is processed and interpreted in a way that perpetuates a sense of serious ongoing threat. Thus, the model emphasises subjective interpretations of the event and its sequelae as an important determinant of post-trauma outcomes. Existing theory suggests subjective peritraumatic responses contribute to the development of PTSD because they influence trauma-related appraisals and the nature of the trauma memory. People who experience strong peritraumatic responses may be more likely to interpret this as evidence for a negative view of themselves (e.g., "I cannot cope with stress"), creating an internal threat to their sense of self as capable and worthy. Additionally, strong peritraumatic responses may prompt data-driven processing (processing of sensory/perceptual characteristics of the event), rather than conceptual processing (processing of the event's meaning) during traumatic events, which may interfere with the encoding of the event and result in trauma memories that are poorly integrated with the wider autobiographical memory network. Poorly integrated trauma memories are thought to contribute to a sense of ongoing threat because they underlie the development of the re-experiencing symptoms (e.g., flashbacks) characteristic of PTSD (Ehlers & Clark, 2000). Biological models suggest peritraumatic distress contributes to increased risk for PTSD through fear conditioning. Strong peritraumatic reactions elicit biological fear responses, resulting in the formation of strong associations between trauma cues and fear responding (Pitman et al., 2012). Continued reactivation of the fear response through exposure to trauma-related cues may lead to sensitization of the neurocircuitry implicated in fear responding, which may contribute to increased risk for PTSD (Perusini et al., 2016).

Findings of the current study are also consistent with existing empirical research. Meta-analyses indicate strong peritraumatic responses are associated with the development of PTSD (Ozer et al., 2003; Vance et al., 2018). Importantly, peritraumatic distress has been found to be uniquely associated with PTSD symptoms in studies controlling for degree of trauma exposure (Cénat & Derivois, 2014; Shigemura et al., 2014), suggesting the association between subjective peritraumatic responses

and PTSD symptoms persists among people exposed to similar degrees of objective exposure. Prior studies of the Canterbury earthquakes have also found subjective responses to be an important determinant of post-trauma outcomes, consistent with the current findings. Boden et al. (2015) investigated the associations between degree of earthquake exposure, peritraumatic distress, distress caused by ongoing earthquake-related disruption, and PTSD symptoms among a birth cohort exposed to the Canterbury earthquakes. Using structural equation modelling, peritraumatic distress and disruption distress were found to fully mediate the association between degree of earthquake exposure and PTSD symptoms, suggesting subjective emotional experiences, both during the earthquakes and in response to ongoing disruption, are important factors contributing to the development of PTSD symptoms. Furthermore, these findings were robust to the inclusion of a range of covariates, including gender, ethnicity, prior mental disorders, and intellectual functioning. Peritraumatic distress was also found to be an important determinant of post-earthquake depression symptoms in a similar study of the same participants (Bell et al., 2017), suggesting peritraumatic distress influences mental health outcomes beyond PTSD symptoms.

Some evidence suggests recall of peritraumatic responses is influenced by current symptom levels; those with PTSD may attribute their symptoms to their experience of the traumatic event, resulting in overestimates of peritraumatic distress (Candel & Merckelbach, 2004). Thus, it is possible this finding reflects group membership, rather than the role of peritraumatic distress in the development of post-trauma outcomes. However, other evidence suggests memories of traumatic events are recalled reliably over time, even more so than memories for positive events, and that reliability is not influenced by the presence of PTSD (Porter & Peace, 2007).

Although a large body of research supports the association between peritraumatic responses and the development of PTSD, the evidence is not unequivocal. For example, van der Velden and Wittman (2008) conducted a systematic review of prospective studies examining the association between peritraumatic dissociation and the development of subsequent PTSD symptoms. Peritraumatic dissociation occurs when an individual feels detached from their surroundings or sense of self during a traumatic event, and has been linked to the development of poorer post-trauma outcomes (Lensvelt-Mulders et al., 2008). Although several studies reported statistically significant bivariate

associations, few studies found peritraumatic dissociation to be an independent predictor of subsequent PTSD symptoms when controlling for other factors, suggesting peritraumatic dissociation has little independent predictive value. Similarly, diagnostic criteria for PTSD no longer include subjective responses to qualify events as traumatic. In DSM-IV, Criterion A2 specified that the individual must have responded to the traumatic event with intense fear, helplessness, or horror (American Psychiatric Association, 2000). This criterion was removed in DSM-5, partly due to research suggesting the addition of subjective peritraumatic responses (Criterion A2) does not increase predictive utility for PTSD beyond endorsement of experiencing a traumatic event (Criterion A1; Bedard-Gilligan & Zoellner, 2008). Evidence suggests there are distinct peritraumatic processes that could be differentially associated with PTSD symptom clusters. Massazza et al. (2021) investigated the factor structure of items from six measures of peritraumatic responding and identified five underlying factors including mental defeat, somatoform dissociation, cognitive overload, immobility, and distress. Different factors were significant predictors for different symptoms clusters, which could partially account for the mixed findings regarding the role of peritraumatic responses in the development of PTSD.

Unlike peritraumatic distress, exposure-related distress (as assessed by the TESS) was significantly associated with group membership in univariate, but not multivariate, analyses in the current study. This finding suggests exposure-related distress was not independently associated with post-trauma outcome when accounting for peritraumatic distress and education level. The measure of exposure-related distress used in the current study captured distress in response to acute experiences (e.g., sustaining injury, being buried under rubble), but also distress associated with ongoing earthquake-related disruption (e.g., damage to home, loss of employment). Thus, this finding is somewhat inconsistent with prior studies identifying subjective responses to ongoing disruption, in addition to peritraumatic responses, as being uniquely associated with PTSD symptoms (e.g., Boden et al., 2015). Findings suggest that while treatment-seeking participants experienced greater levels of exposure-related distress compared to control participants (who coped well), subjective peritraumatic responses were relatively more important in determining post-trauma outcomes.

Education level also emerged as an important factor distinguishing treatment-seeking and control participants; treatment-seeking participants were less likely to

have a tertiary educational qualification compared to those who coped well. Meta-analyses have identified lower education level as a risk factor for PTSD among adults following earthquakes, consistent with the current findings (Tang et al., 2017). However, the high proportion of participants with a tertiary educational qualification in the control group (76%) may reflect the convenience sampling strategy used to recruit these participants. Lower education level may act as a proxy for lower socioeconomic status. Individuals of lower socioeconomic status may have fewer resources (e.g., financial resources) available to help cope with ongoing earthquake-related stressors (e.g., property damage), which may increase risk for the development of PTSD symptoms (Hobfoll et al., 2006).

Number of pre-earthquake mental disorders, number of life events experienced in the past 5 years, and difficulty associated with life events, as well as the remaining sociodemographic factors of interest (gender, age, and relationship status) were significantly associated with post-trauma outcome in univariate analyses but did not uniquely distinguish participants in multivariate models. This is inconsistent with prior studies in which factors relating to pre-trauma mental disorders (Ogle et al., 2016), life stress (Bonanno et al., 2007), and sociodemographic characteristics (particularly female gender; Wang et al., 2011) have been identified as unique predictors of PTSD symptoms. However, it is difficult to make direct comparisons between these findings and the current findings, as prior studies have controlled for different sets of variables in multivariate analyses. Of note, studies reporting unique associations between pre-trauma mental disorder and life stress and PTSD symptoms have not controlled for peritraumatic distress (e.g., Bonanno et al., 2007; Ogle et al., 2016), which could partially account for discrepancies between the current findings and those of prior studies. Peritraumatic distress may be an important factor to control for statistically in future studies examining factors associated with post-trauma outcomes.

Results of the current study have important potential clinical applications. Peritraumatic distress and education level emerged as important factors distinguishing treatment-seeking and control participants with moderate to high degrees of earthquake exposure, suggesting these factors would be useful in screening for vulnerable individuals following potentially traumatic events. Screening for subjective peritraumatic responses could be particularly important in the context of disasters, which often result in large numbers of people with high

levels of objective exposure. Screening for subjective peritraumatic responses has potential to improve the accuracy of screening efforts and ensure that resources, which are often limited in post-disaster periods, are used efficiently.

The current study should be interpreted considering the following limitations. Treatment-seeking participants were drawn from a clinical service population and may not be representative of all Canterbury residents who sought treatment for PTSD following the earthquakes. Although the service was the main provider of publicly-funded mental health treatment for those with severe earthquake-related distress, some people received treatment through other services (e.g., private psychologists and psychiatrists). Furthermore, treatment-seeking participants may not be representative of all individuals who developed PTSD after the earthquakes, given some people are likely to have not sought treatment at all. For instance, some research indicates men are less likely to seek treatment following trauma exposure than women (Gavrilovic et al., 2005), suggesting women may have been overrepresented in the treatment-seeking sample. Additionally, a convenience sampling strategy was used to recruit control participants, so these participants may not have been demographically representative of all Canterbury residents who coped well following the earthquakes. For instance, the control group was characterised by a large proportion of people with a tertiary educational qualification, which may have partially accounted for the emergence of education level as an important distinguishing factor. Additionally, New Zealand European participants were overrepresented in both the treatment-seeking and control groups. Thus, results here should be interpreted with caution. Future studies are needed to determine the generalisability of these findings to other populations, and could clarify why some established predictors of post-trauma outcome did not emerge as significant distinguishing factors in the current study. The current study was cross-sectional. Post-trauma trajectories are dynamic processes and cross-sectional studies may fail to capture important changes occurring over time. The use of retrospective self-reports may have compromised the accuracy of some self-report measures, such as peritraumatic distress, and pre-earthquake MINI disorders, which were based on post-earthquake recall. Some factors previously found to be associated with post-trauma outcomes (e.g., social support, prior trauma exposure) were not assessed in this study and thus were not included in multivariate models. Some participants did not have available data for all variables, which may have

reduced power to detect effects.

The current study also has considerable strengths. We compared participants who experienced markedly different outcomes following similar degrees of objective exposure to the same event, allowing for an examination of factors associated with post-trauma outcomes among people with similar objective experiences. Control participants were recruited into the study only if they had experienced substantial earthquake exposure, ensuring these participants were not coping well simply due to lower levels of exposure severity. Clinician-administered interviews were used to assess the presence of current and pre-earthquake mental disorders, ensuring this information was of high quality. Well-validated measures were used to assess participants' current psychological functioning.

To conclude, the current study identified factors distinguishing treatment-seeking individuals from those who did not develop any mental health issues and self-identified as having coped well following a major earthquake. Findings suggested that while the groups experienced comparable degrees of objective earthquake exposure, treatment-seeking participants reported higher levels of peritraumatic distress and lower levels of education, which potentially contributed to the development of distinct post-earthquake outcomes. The findings here suggest subjective peritraumatic responses and education level could be included in screening efforts following disasters.

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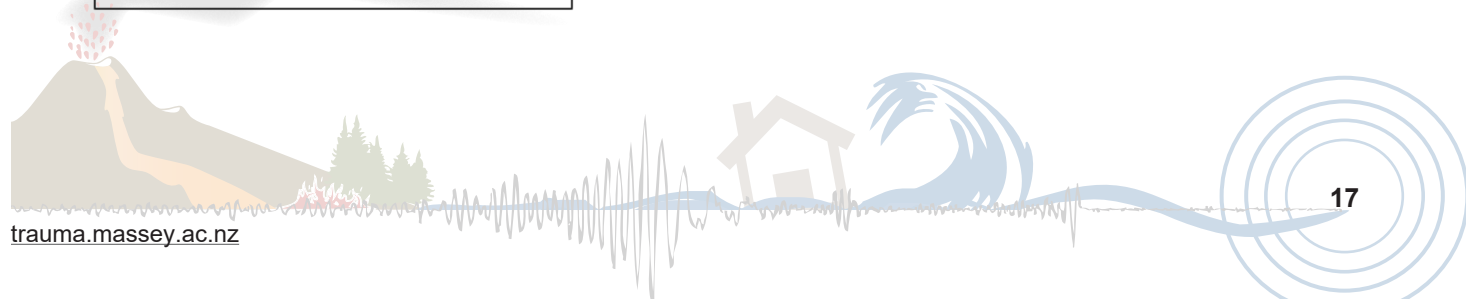
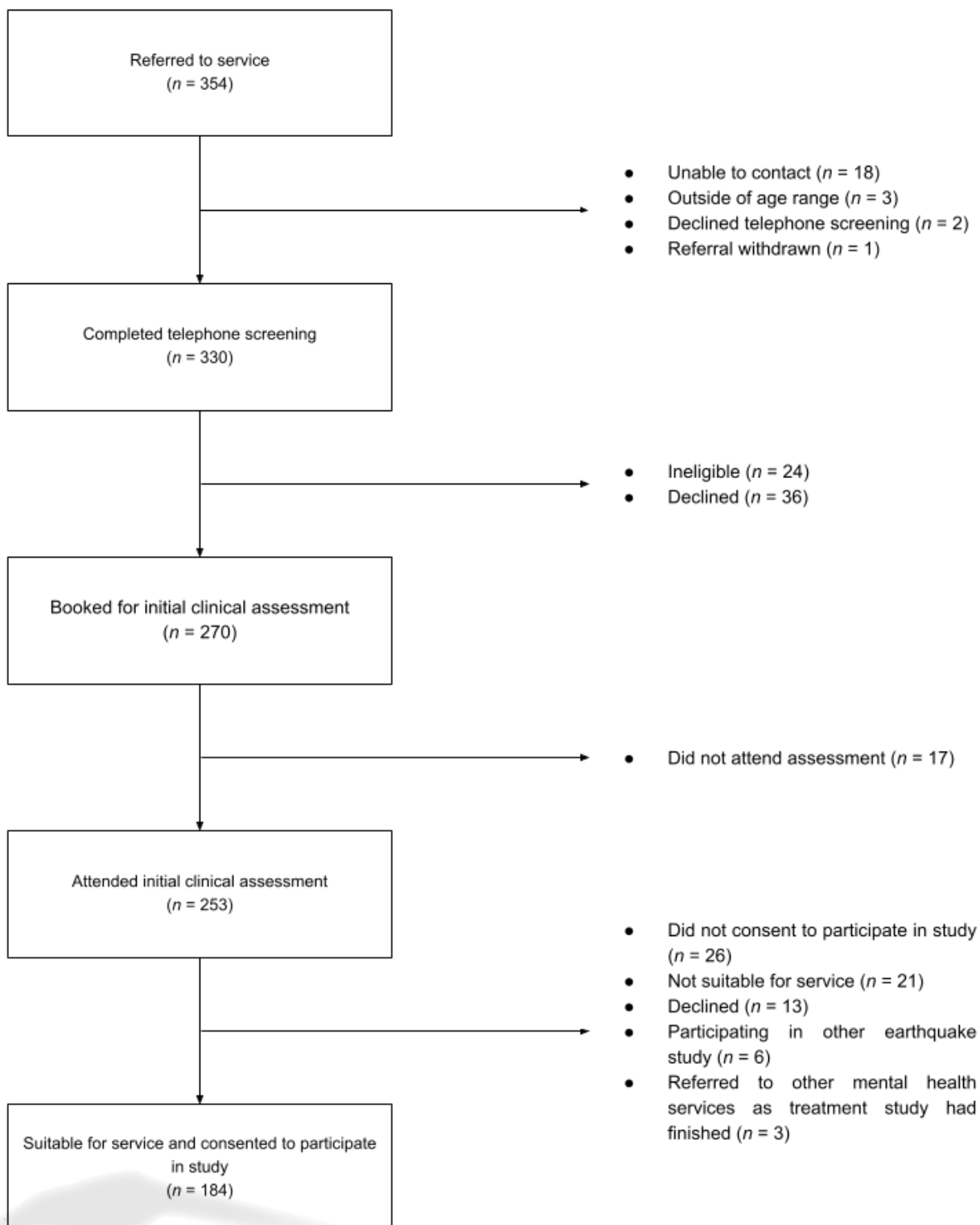
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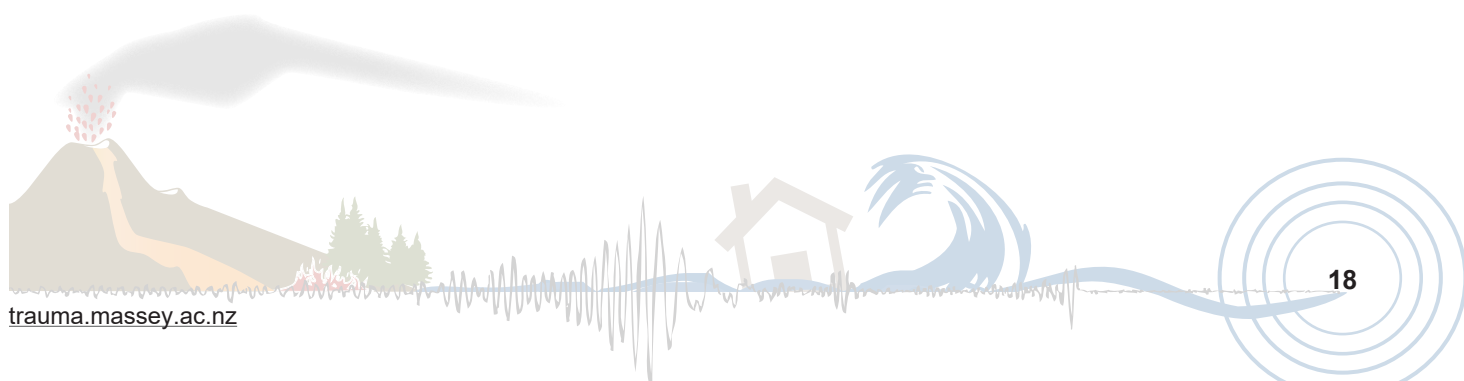
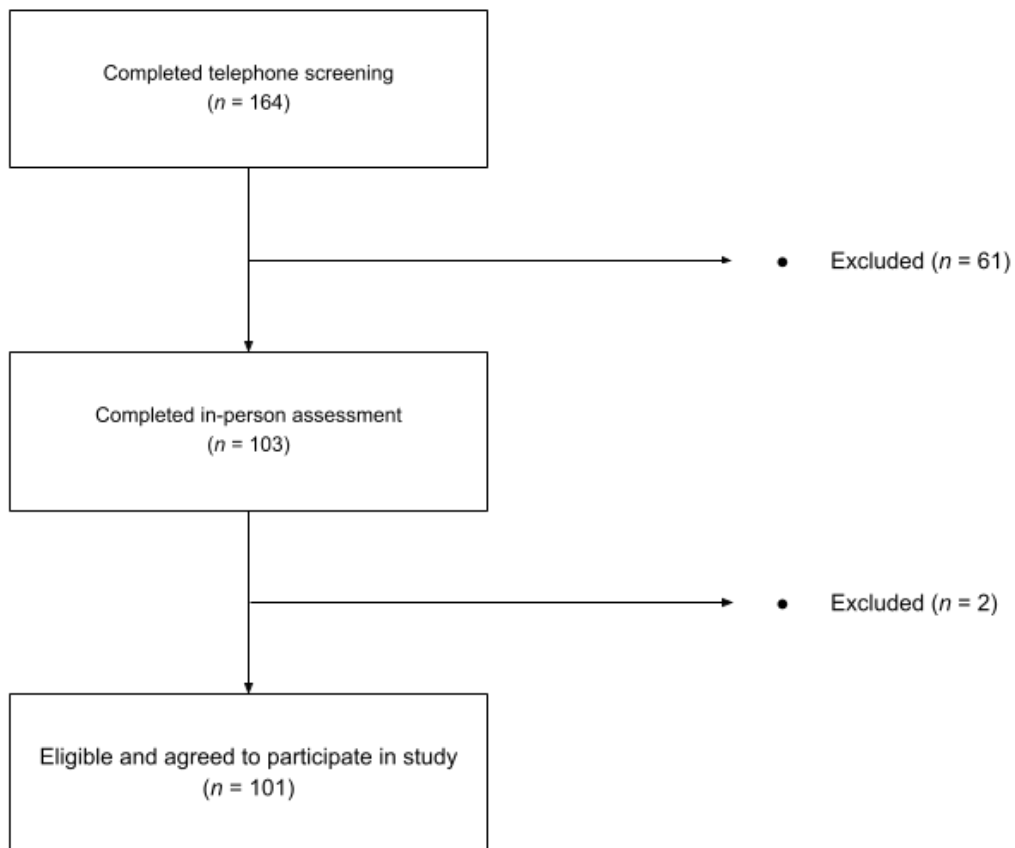
Appendix A

Flow of Treatment-Seeking Participants Through the Recruitment Process



Appendix B

Flow of Control Participants Through the Recruitment Process



An educational activity on climate change for secondary students in Vanuatu: Does gender influence performance?

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Abstract

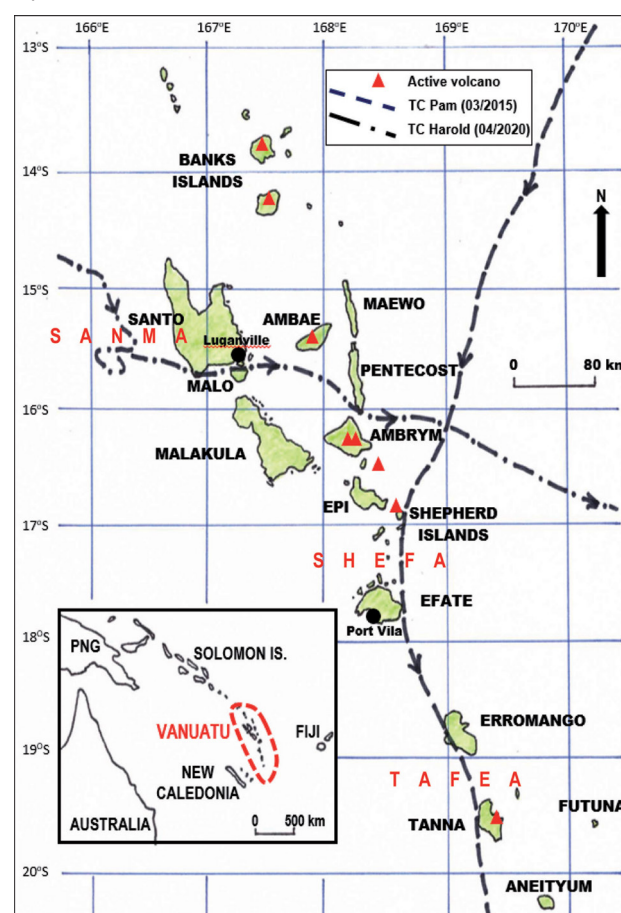
As part of wider research into the effectiveness of formal and traditional learning about climate and disaster resilience in the Pacific Island nation of Vanuatu, a mixed-method investigation was conducted during 2020 into the effectiveness of a pictorial resource depicting key elements of climate change education used in a discovery learning activity at junior secondary school level. Students completed a questionnaire before and after the intervention so that changes in their knowledge, attitudes, and behaviour could be measured. The quantitative data obtained was supplemented by qualitative information from teachers on the reasons for patterns observed. Results were compared across schools, age level, language of instruction, gender, and urban/rural location; this paper focuses on comparisons by gender obtained from a sample of 209 students, most of whom were in Years 9-10. While average overall scores for boys and girls, both before and after the intervention, were almost identical, girls performed better when their teacher was female and boys when their teacher was male. Also, girls had significantly higher scores than boys when the activity was conducted in urban schools, but not in rural schools. Reasons for this difference include a tendency for boys to be more easily distracted from their studies than girls in an urban environment.

Keywords: Vanuatu, climate change, gender, discovery learning, secondary education, junior secondary level, climate resilience, climate change adaptation, mixed methods

Vanuatu, an archipelago of over 80 volcanic islands in the south-west Pacific (Figure 1), is highly vulnerable to the impacts of climate change and disasters (Connell, 2013; Le Dé et al., 2018). Vanuatu's location in the tropical Pacific makes it susceptible to cyclones, floods, droughts, landslides, and sea-level rise, while its position on the "Ring of Fire" exposes it to earthquakes, volcanic eruptions and tsunamis (Pierce, 2019; Walshe & Nunn, 2012).

Figure 1

Map of Vanuatu



Note. Source: Author.

The 2021 World Risk Report, calculated for 181 countries on the basis of exposure and vulnerability (susceptibility, lack of coping capacities, and lack of adaptation capacities), identifies Vanuatu as the planet's most at-risk country to natural hazards, with an index of 47.73 on a range of 0-100 – considerably higher than that of the next two countries, the Solomon Islands (31.16) and Tonga (30.51; Bündnis Entwicklung Hilft, 2021).

In the 2020 national census, Vanuatu had a population of 300,019, of whom 66,753 (22%) were living in the two main urban areas of Port Vila and Luganville (Vanuatu National Statistics Office, 2021). In the same year, Vanuatu's Ministry of Education and Training (MOET) reported that total school enrolment was 96,829, of whom 23,943 were at secondary level (Years 7 to 14, with an official age range of 12 to 19; MOET, 2021).

The broader research project, conducted between October 2019 and November 2021, had the question "How effective is formal education on climate and disaster resilience in Vanuatu in terms of knowledge and skills gained, changes in attitude and behaviour and impacts on individuals and their communities?" (Pierce, 2023). This was applied to four academic levels – primary, junior secondary (Years 7 to 10), senior secondary (Years 11 to 13/14), and post-secondary. I investigated the impact of curricular materials and pedagogy on student learning about climate change and disasters, seeking to determine the level of resilience achieved. It must be stressed that this research in formal education was conducted in order to assess the impact of specific educational programmes or courses on the students involved, and not to assess the students themselves. The students served as agents for measuring the effectiveness of those courses or educational resources.

At junior secondary level, the existing Vanuatu curricula in Science and Social Science had minimal content on climate change or disaster risk reduction. New curricula were being developed but were not ready for use. The revision would mean that new topics related to climate change, disasters, and resilience would be inserted into syllabi for Social Science and Science, with the revisions linked to the United Nations Sustainable Development Goals 1-17, particularly Goal 13 on Climate Action (United Nations Department of Economic and Social Affairs, 2015).

There was, however, an educational resource for Pacific Island schools known as "Learning about Climate Change the Pacific Way" that had been developed between 2011 and 2013 in order to address the gap in learning materials

at junior secondary level on resilience to climate change and disasters in a Pacific Island context. It was one of the few resources available to schoolchildren and their teachers in Vanuatu in relation to resilience. It was produced under the auspices of the Secretariat of the Pacific Community (SPC) for use in five Pacific Island nations – Fiji, Kiribati, Samoa, Tonga, and Vanuatu – with funding provided by the SPC and Deutsche Gesellschaft für International Zusammenarbeit (GIZ) through their Coping with Climate Change in the Pacific Island Region Programme. Designed for Years 9 and 10, this "Climate Change Toolkit" comprises 16 large pictures for classroom display that deliver key messages relevant to climate change science, impacts in the Pacific, and options for mitigation and adaptation (SPC & GIZ, 2014). The images depict: the water cycle; climate globally and in the Pacific; El Niño and La Niña; causes of climate change; air-land-ocean interrelationships; the changing climate of Vanuatu; impacts of cyclones and droughts; mitigation and adaptation activities; adaptation strategies for cultivation, livestock, forestry, agro-forestry, and fishing; adaptation in urban areas; and an overview of all strategies on a Pacific island before and after adaptation. Figure 2 provides an example of one of these images.

Figure 2

Picture 11 of the Climate Change Toolkit and its Associated Questions



1. What actions can we take to reduce the levels of GHGs in the atmosphere? (Called "mitigation")
2. What actions can be taken to change our way of life so that it fits a climate that is warmer, with stronger cyclones and more droughts? (Called "adaptation")
3. What are the actions shown in the middle of the diagram?

Note. Source: SPC and GIZ (2014).

The Toolkit is accompanied by a comprehensive Teacher's Guide that provides background information and practical classroom activities applicable to both junior and senior levels of secondary education (SPC & GIZ, 2013). Its use was briefly critiqued by Walid (2017), but otherwise there have been no other attempts to evaluate its effectiveness. Walid's main criticisms were that:

Although the curriculum accounts for the power of tradition and uses pictures to communicate ideas directly, it omits traditional pedagogical methods of communication, such as story-telling. The project also does not cover all grades. Additionally, as this project relies heavily on top-down communication through government documents, it does not consider equally voices from the local communities.

(Walid, 2017, p. 277)

Sets of the Toolkit were distributed by Vanuatu's Curriculum Development Unit to junior secondary schools throughout the nation during 2013-2014, with short training courses for teachers on its deployment. By 2019, however, enquiries revealed that it had been largely forgotten. One reason given was the widespread destruction of learning materials as a result of Category 5 Tropical Cyclone Pam in March 2015; another was the frequent transfer of teachers between schools, with those who had received training moved into new posts and often asked to teach different subjects.

For my research into the effectiveness of learning about climate and disaster resilience at junior secondary level, I therefore set out to examine whether this resource does have intrinsic value in changing students' understanding of climate change and disasters, as well as attitudes and pro-environmental behaviours. At the same time, the aim was to increase awareness among new and existing teachers of the resource's existence, and how it might advance learning in the current absence of other materials on climate and disaster resilience for students at junior secondary level. I wanted to examine the impact of this educational resource on students in a variety of settings, as well as looking at variables that might influence its effectiveness: for example, language of instruction, urban or rural location, year level, and gender.

Literature Review

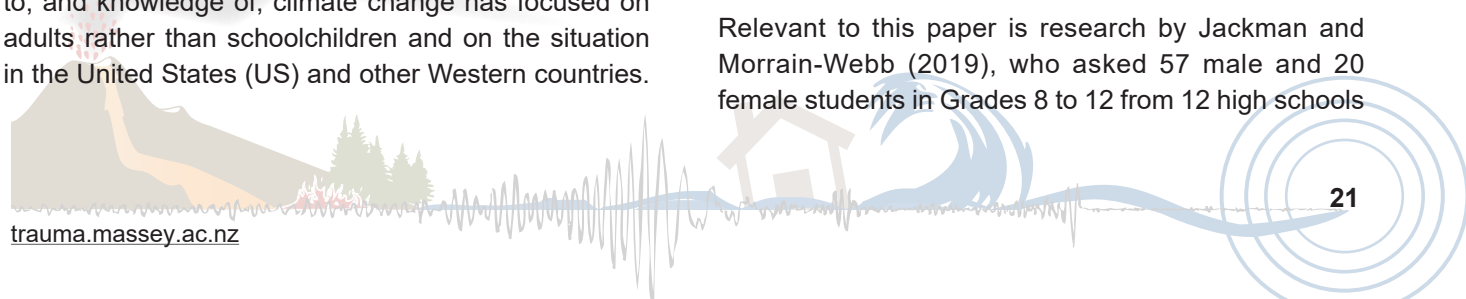
Gender differences among adults regarding climate change. Literature on gender differences in attitudes to, and knowledge of, climate change has focused on adults rather than schoolchildren and on the situation in the United States (US) and other Western countries.

For example, women are typically more likely than men to believe that climate change is happening (Hornsey et al., 2016; McCright et al., 2013), worry about its effects (McCright, 2010; McCright & Sundström, 2013), perceive more climate risks (Brody et al., 2008; Hamilton, 2011; van der Linden, 2015), express more knowledge about climate change (McCright, 2010), and perceive global warming as posing a threat within their lifetime (Hamilton, 2011). Women are also more likely than men in several economically wealthier countries (US, Canada, United Kingdom, Australia, and Italy) to endorse the view that people will need to make major changes in their lifestyles to reduce the effects of climate change. Women's greater environmental concerns are thought to result from their heightened risk perception (Bord & O'Connor, 1997; McCright, 2010; Xiao & McCright, 2012) and their greater propensity to show compassion and express an "ethic of care" (Zelezny et al., 2000, p. 445). Ballew et al. (2018) found that although similar proportions of men and women think that global warming is caused by humans, women more strongly believe that global warming will harm them personally and will harm people, plants, and animals as well as future generations. On the other hand, women were less likely than men to know the scientific facts about global warming and expressed more uncertainty about their knowledge.

Note that all the above studies relate to adult perceptions of climate change gained through public communications, as opposed to perceptions of school students gained in the classroom through climate change education.

Differences between boys and girls in academic performance at secondary school level. The influence of gender on the academic performance of school-age children is a subject of long-standing debate. Hyde (2004) proposed the "gender similarities hypothesis", arguing that boys and girls are more alike than they are different. On the other hand, research in both economically developed and developing countries indicates that since the end of the 20th century, girls are outperforming boys at secondary level (Hadjar et al., 2014; Jha & Kelleher, 2006; Nnamami & Oyibe, 2016; Ullah & Ullah, 2019; Van Houtte, 2004). In North Pentecost, Vanuatu, Mahuri (2019) found that certain socio-cultural practices jeopardize both participation (school attendance) and performance (academic achievement) of all schoolchildren, with boys impacted to a greater extent than girls.

Relevant to this paper is research by Jackman and Morrain-Webb (2019), who asked 57 male and 20 female students in Grades 8 to 12 from 12 high schools



in Trinidad and Tobago why they felt that girls were outperforming boys in high school. Among the reasons given, the most common was that females' greater success is due to the current societal thrust towards female empowerment and gender equity, which aids girls' academic achievement but can also immobilize boys. A second reason was the effects of negative influences in society that impact more on boys than on girls: gang violence, delinquency, criminal activity, illicit drug use, and abuse. A third reason was that male students focus on "extracurricular activity" (especially sport), rather than on schoolwork. A fourth reason related to a belief that girls are more goal-focused, emotionally mature, and have a better work ethic. Other factors mentioned were that the education system favoured females – linked to the "feminization of schooling" hypothesis (Figueroa, 2000; Joseph, 2016) – and that there is a lack of parental direction for boys.

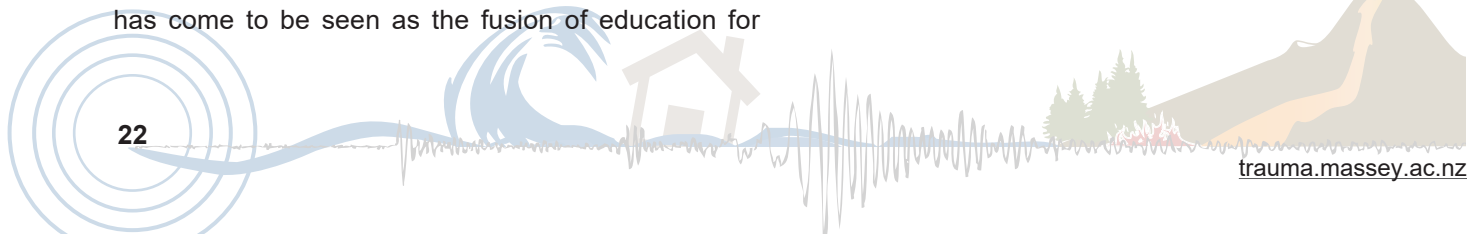
As we grow older, the reorganization of brain connections may explain why females generally mature faster in certain cognitive and emotional areas than males during childhood and adolescence (Lim et al., 2013). There appear to be general differences in brain development between male and female cohorts, and that modern pressures make it difficult for the young human brain to form optimal connections during adolescence, with boys being especially vulnerable (Bergland, 2013). These findings resonate with those of Brouwer et al. (2021), who investigated differences in brain age between males and females and found that although females and males show similar developmental ("aging") patterns, on average females pass through this development earlier, being ahead of males by at most a year. We may possibly infer that cognitive development, or the growth of a child's ability to think and reason (Stanford Medicine, 2024), occurs earlier in adolescent girls than it does for adolescent boys, and that this may impact on their faster academic progress in schools.

Differences between boys and girls regarding climate change education at secondary level. Most literature on climate change education appeared after the United Nations Educational Scientific and Cultural Organisation (UNESCO) launched the Decade of Education for Sustainable Development in 2010. Increasingly, greater focus has been placed on the role of adaptation in building resilience to impacts, with the aim of empowering learners to learn by doing and take action for sustainable development (Stevenson et al., 2017; UNESCO, 2015). Moreover, resilience education has come to be seen as the fusion of education for

climate change with education for disaster risk reduction (Kagawa & Selby, 2012), particularly for Pacific Islands, where climate change is viewed as a slow-acting disaster (Hemstock et al., 2017). It is within this context that the Toolkit is of relevance, since it has a particular emphasis on building adaptive capacity; according to the Teacher's Guide for this resource, the focus of learning is on "the effects of changes in air and sea-surface temperature, rainfall, sea-level rise and extreme weather events on environments, economies and people" and the need to "enhance individual and community skills to adapt to these changes" (SPC & GIZ, 2013, p.1).

Literature on the particular influence of gender on students' performance in climate change education at secondary school level is limited. Relevant to the present research is a study in Tanzania, where Kira and Komba (2015) collected information from 480 secondary school students in Morogoro region and found that boys were significantly more knowledgeable than girls in their understanding of climate change; they had greater awareness, for example, of the role played by carbon dioxide in global warming and of industries and automobiles as the most important contributors to greenhouse gas emissions.

This paper aims to address the gap in research on the role of gender in student performance in climate change education, particularly in terms of adaptation and resilience. We explore whether participation in the Toolkit activity by students in Vanuatu yields a differentiation in performance by gender. In this context, "performance" will be measured by scores achieved in a "before/after" diagnostic questionnaire for knowledge, attitudes, intended patterns of behaviour, and all three aspects together. This paper also seeks to determine whether teacher gender is a factor in educational success. For example, Muralidharan and Sheth (2016) found that male and female teachers of Grades 1-5 in Andhra Pradesh, India, were more effective in teaching students of their own gender, but that female teachers had greater effectiveness overall. However, Nnamani and Oyibe (2016), researching secondary students of Social Studies in Nigeria, discovered that male and female students taught by male teachers obtained higher mean scores than male and female students taught by female teachers.



Methods

Use of Quantitative Tools

This research used quantitative methods to determine the effectiveness of the Toolkit in changing students' knowledge, attitudes, and intended behaviour, and hence their level of resilience to climate change and disasters. A diagnostic questionnaire was designed for use before and after students undertook the Toolkit activity and this was to be completed by students at Year 9 and 10 level in a sample of classes selected from the 111 secondary schools in Vanuatu.

Selecting the Sample

This sample was not selected on a random basis. Instead, I relied on my long-standing contacts with teachers, especially those whom I had taught or trained over four decades of service in the country – teaching at secondary level, in tertiary level courses at USP's Emalus Campus in Port Vila, and at the Vanuatu Institute of Teacher Education (VITE). I identified possible English- and French-medium schools in urban and rural settings on several islands, then consulted with school principals and teachers to find classes whose teachers were willing to participate in the research. In this way, a sample of 19 classes was selected, with all students in those classes completing the same questionnaire both before and after undertaking the Toolkit activity. Although the focus was on Year 9-10 students, a few teachers felt that the activity would help students studying climate change in their Earth Science and Geography courses at senior secondary level, and asked to use it as a supplementary resource for those students. This was deemed acceptable as it could later enable a comparison of performance by year level. Questionnaires were completed by a total of 363 students from 19 classes at Year 9-12 level, of whom 296 (82%) were in Years 9 or 10.

Thus the sampling methods used for evaluating students' participation in resilience education through the Toolkit activity are complex. Schools and teachers were selected through non-probability sampling. The researcher used methods that fit the definitions of convenience and purposeful sampling (DeCarlo, 2018; Marshall, 1996), making deliberate choices to ensure participation from a range of schools in different settings. But once the schools and classes had been selected, then all students in each class were involved. All classes in Vanuatu schools are mixed, and at secondary level, females slightly outnumber males: in 2020, 12,287 females were enrolled in Years 7 to 14 as compared with 11,656 males (MOET, 2021). Those in Year 9 and above have already

passed a national examination at the end of Year 8, so that broadly speaking they are of similar academic ability, even though they vary in terms of family background, culture, and gender. In that sense they constitute a uniform population regardless of the school they attend, since in my experience, the Vanuatu school system is relatively equitable.

Investigating the Influence of Gender

The selection of schools and classes was made through purposeful sampling in order to capture a balance between urban and rural schools and between classes taught by female and male teachers, as well as similar overall numbers of boys and girls. Overall, data was disaggregated by gender for 112 girls and 97 boys from eight classes in five schools. The classes taught by female teachers had 74 female and 53 male students, while those taught by male teachers comprised 38 females and 44 males. Thus in terms of participation, classes with female teachers had higher proportions of females than males, while classes with male teachers had higher proportions of males than females. This may simply be co-incidence, or it may reflect my own observation from 35 years of teaching in Vanuatu that teenage boys at Years 9-10 level are more likely than girls to miss classes when their teacher is female. Note that 84% of Year 9 students are aged 14-17, and 85% of Year 10 students are aged 15-18 (MOET, 2022, calculated from Table 18).

Qualitative Tools

In order to explore potential reasons for differences in the performance of girls and boys, and between girls and boys in rural and urban settings, I conducted email interviews with seven of the teachers involved and three secondary school principals. In this way, a qualitative tool supplemented the quantitative data already obtained, triangulating information obtained from students with that gained from teachers.

Ensuring a Uniform Pedagogical Approach to the Toolkit Activity

In order to gain a realistic assessment of the Toolkit's effectiveness, I needed to ensure that the selected teachers would use the Toolkit activity in a uniform pedagogical manner. For this reason, I conducted a series of face-to-face training sessions in English and French between December 2019 and March 2020 for 17 teachers of Basic Science, Social Science, Geography, and Earth Science in each of six secondary schools, as well as for 13 teacher trainees graduating from the

VITE. I encouraged teachers to use the pictures in a discovery learning activity whereby the 16 images were pinned up in the classroom and students collaborated in pairs or small groups to visit each picture and answer its associated questions. This discovery learning activity might extend over 1 to 3 hour-long classroom sessions, depending on the students and the teacher, concluding with general whole-class discussion in which teacher and students would together explore and clarify key concepts. Figure 3 shows an example of one such teacher training session, while Figure 4 shows a group of students carrying out the activity.

Figure 3

Teachers of Year 10 Social Science at Nagavika College Participate in Training on the Climate Change Toolkit



Note. Here they are responding to questions on each picture as though they are students. Source: Author.

Figure 4

A Group of Students Discussing Picture 2



Note. Source: Social Science Teacher, Acacia Secondary School.

In order to assess changes in students' knowledge, attitudes, and behaviours as a result of this intervention, teachers would ask each individual student to complete the diagnostic questionnaire before undertaking the activity and then again at its conclusion (Figure 5), after the class had discussed the pictures with their teacher. Questions in the questionnaire were not the same as those accompanying each picture but attempted to draw out key aspects of learning generated by the Toolkit as a whole.

Figure 5

Individual Students Completing the "After" Questionnaire



Note. Source: Social Science Teacher, Pandanus Junior Secondary School

In all, 363 students from 19 classes in nine schools participated in the activity between March 2020 and March 2021. Implementation of the Toolkit activity was delayed by the declaration of a State of Emergency (SOE) in Vanuatu in April 2020 due to the impacts of Category 5 Cyclone Harold and the start of the COVID-19 pandemic. Restrictions on international travel meant that once I left Vanuatu in March 2020, I was unable to return until mid-2022. Progress was therefore monitored through virtual means. Completed "before" and "after" questionnaires were transmitted electronically, and interviews with teachers conducted through emails, texts, or phone.

Data Collection and Analysis

A mixed methods approach to the research was followed, with quantitative analysis of students' responses supported by qualitative data obtained from class teachers and school principals.

For the quantitative data, each class teacher asked his/her students to complete the diagnostic questionnaire before and after the intervention, then send all completed forms to the researcher. Questionnaires were marked according to the scheme shown in the Appendix, then the results for each class were tabulated. The diagnostic questionnaire comprises 20 agree/disagree (A/D) statements, each worth one point, and seven short-answer questions, each marked on a four-point scale according to levels of displayed competence. In marking the A/D statements, +1 was awarded for a correct answer, -1 for an incorrect answer and 0 for "Don't know", with this scale adopted in order to differentiate correct from incorrect responses. An exception was made for statement 18, where a respondent was not penalized for disagreeing to participate in a student strike for climate change because there are other ways in which concern can be expressed. For the longer answers required in questions 21 to 27, marking followed descriptors at four

levels (accurate, satisfactory, limited, and irrelevant/no answer).

For purposes of analysis, items 1-16 and 21-26 in the questionnaire were considered as indicators of knowledge, items 17-20 of attitudes/opinions, and item 27 of intended behaviour. Statement 19 was considered as an attitude as it reflects a student's level of confidence in his/her own ability. Scores can range from -1.0 to +1.0 for items 1 to 20, and from 0.0 to +5.0 for items 21 to 27. The higher the score, the higher the degree of resilience implied and so the better the performance.

For purposes of confidentiality, schools were given anonymous names that correspond to common tree species in Vanuatu.

Results

Comparison of Male and Female Scores: Averages for all Items, Knowledge, Attitudes, and Behaviour

Table 1 and Figure 6 show a comparison of average scores for females and males before and after the intervention. For all items, scores were almost identical: 0.49 for both genders before the activity and 0.81 for females and 0.80 for males afterwards.

For knowledge items, there was a similar pattern, with girls attaining average scores of 0.44 beforehand and 0.72 afterwards, with an increase of 0.28, while the comparable scores for boys were 0.45 and 0.71 with an increase of 0.27. For attitudes, females had a higher average score than males before the intervention (0.58 to 0.48), but a lower score afterwards (0.73 to 0.78), so that the improvement for females was half that for males (0.15 to 0.31). In terms of behaviour, males had a higher score than females beforehand (1.42 to 1.25), and a lower score afterwards (2.69 to 3.08), with a lower degree of improvement (1.27 to 1.83). Note that the scores for behaviour are based on one item only (Q27), which has a mark range of 0 to 5.

If, however, the average scores by gender are disaggregated into responses to the 20 A/D items and responses to the seven short-answer questions (Figure 7), a different pattern emerges for the knowledge items. Scores for the A/D items were slightly higher for boys than girls before the intervention (0.38 to 0.35) and afterwards (0.56 to 0.54), while it was the reverse for the short answer questions, with higher scores for girls beforehand (0.72 to 0.62) and again afterwards (1.16 to 1.10). The Toolkit seems to have improved girls' performance in the short answer questions for both

knowledge and behaviour to a greater extent than for boys, meaning that an understanding of climate change and its impacts and solutions had increased more for girls than it had for boys.

Figure 6

Comparison of Average Scores for Females and Males Before and After the Intervention

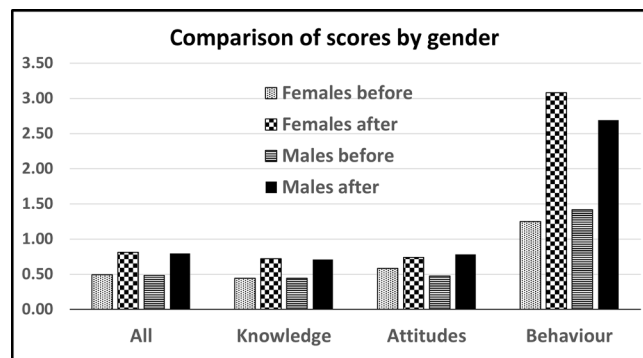
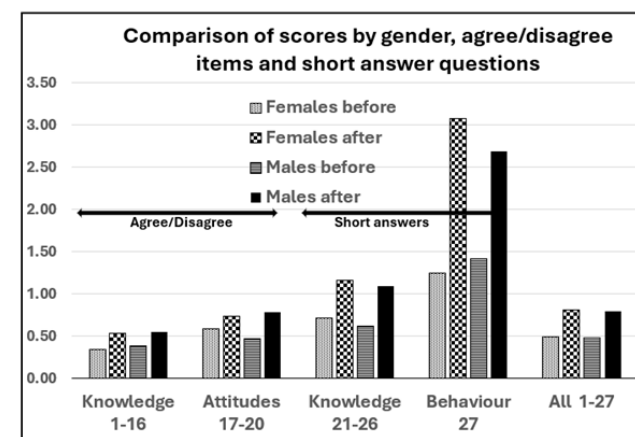


Figure 7

Comparison of Average Scores for Females and Males Before and After the Intervention, Differentiated According to Agree/Disagree Items and Short Answer Questions



Comparison of Male and Female Scores for Individual Items

Performance by gender can also be considered on the basis of individual items. Table 1 shows that for the A/D items, exposure to the Toolkit has resulted in boys outperforming girls in nine of the 16 items (Qs 2, 3, 4, 5, 9, 10, 12, 15, 16) that relate to climate science, fishing, and Vanuatu's vulnerability to disasters. Girls outperformed boys in Qs 1, 6, 7, 8, 11, 13, and 14, which refer to the effect of El Niño and definitions of weather, the greenhouse effect, adaptation, and sustainable agriculture. With the short answer questions, the intervention resulted in girls clearly outperforming boys in Qs 21, 26, and 27, which apply to the definition and

Table 1*Comparison of Average Scores for Females and Males Before and After the Intervention*

Statement / Question no.	Females (n = 112)			Males (n = 97)		
	Before Av. score	After Av. score	Change	Before Av. score	After Av. score	Change
1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.	0.94	0.92	-0.02	0.80	0.82	0.02
2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.	0.47	0.53	0.05	0.38	0.65	0.27
3. Our climate changes because of both natural and human factors	0.63	0.85	0.22	0.63	0.92	0.29
4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.	-0.30	-0.30	0.00	-0.04	-0.16	-0.12
5. Climate variability means that the climate of a place may change from year to year.	0.25	0.54	0.29	0.23	0.64	0.41
6. During an El Niño period, Vanuatu experiences droughts.	0.48	0.76	0.28	0.69	0.68	-0.01
7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.	0.30	0.76	0.46	0.33	0.64	0.31
8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.	0.27	0.78	0.51	0.25	0.48	0.24
9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.	0.16	0.43	0.27	0.18	0.46	0.29
10. Carbon dioxide is absorbed by forests and the oceans	0.29	0.45	0.16	0.38	0.77	0.39
11. Coral reefs are being damaged by warmer temperatures and ocean acidification.	0.39	0.68	0.29	0.46	0.66	0.20
12. Vanuatu's greatest climatic dangers are cyclones and droughts.	0.86	0.89	0.04	0.70	0.91	0.21
13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.	-0.27	-0.38	-0.12	0.00	-0.54	-0.54
14. Using compost is a sustainable form of gardening.	0.72	0.92	0.20	0.56	0.80	0.25
15. Catching fish with nets that have very small holes is a sustainable form of fishing.	-0.05	0.10	0.15	-0.01	0.39	0.40
16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.	0.41	0.71	0.30	0.55	0.76	0.22
17. I am worried that climate change will bring great dangers to the world in the future. *	0.63	0.88	0.24	0.42	0.81	0.39
18. I want to join a student strike to show my concern about climate change.*	0.69	0.70	0.01	0.71	0.81	0.10
19. I know a lot about the causes and impacts of climate change.*	0.20	0.44	0.24	0.11	0.61	0.49
20. I want to help my community prepare for climate change and disasters.*	0.82	0.93	0.11	0.66	0.90	0.24
21. What does "climate change" mean?	0.57	0.91	0.34	0.27	0.67	0.40
22. What is the enhanced greenhouse effect?	0.59	0.88	0.29	0.47	0.86	0.38
23. What are 3 human activities putting extra GHGs into the atmosphere?	1.47	2.08	0.61	1.16	2.08	0.92
24. What causes ocean acidification?	0.35	0.73	0.38	0.41	0.72	0.31
25. Why is forestry important for mitigating climate change?	0.66	1.15	0.49	0.72	1.21	0.48
26. What are some of the main impacts of climate change in Vanuatu?	0.58	1.46	0.88	0.67	1.23	0.56
27. What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters?	1.25	3.08	1.83	1.42	2.69	1.27
Knowledge	0.44	0.72	0.28	0.45	0.71	0.27
Attitudes	0.58	0.73	0.15	0.48	0.78	0.31
Behaviour	1.25	3.08	1.83	1.42	2.69	1.27
All	0.49	0.81	0.32	0.49	0.80	0.31

Note. * = Attitude questions; Item 27 = Behaviour question; all other items = Knowledge questions.

impacts of climate change, and behaviours to prepare for climate change and disasters. Girls slightly outperformed boys in Qs 22 and 24, concerning knowledge of the greenhouse effect and ocean acidification. Girls and boys achieved equal scores in Q 23, on anthropogenic causes of climate change, while boys outperformed girls in Q 25, relating to knowledge of how forests mitigate climate change.

Teacher Gender and Student Performance

Turning now to the question of whether the gender of the teacher affects student performance, Table 2 shows results for students in mixed classes taught by female teachers and those taught by male teachers. Average scores for all items before students participated in the Toolkit activity were significantly higher in classes taught by female teachers (0.54) than those taught by male teachers (0.41). However, participation in the activity enabled students taught by male teachers to attain almost the same score (0.79) as those taught by female teachers (0.81). The same pattern can be observed for performance in knowledge items, with classes taught by female teachers having initial scores of 0.50 as compared with 0.36 for classes with male teachers, and then the intervention bringing the scores for male-taught classes to almost the same level (0.71) as that for female-taught classes (0.72). For attitudes, classes taught by female teachers scored more highly before the start of the activity (0.58 to 0.46) but saw a smaller increase (0.73 to 0.80). In the one short answer question on behaviour, male-taught classes had a higher score at the start (1.38 to 1.30) but showed a smaller improvement (2.71 to 3.02), implying that in this respect, female teachers were more effective than males in helping their students become aware of a greater range of adaptation methods.

For individual items, participation in the intervention meant that classes taught by male teachers achieved much higher scores than those taught by females in knowledge items 1, 2, 6, 7, 11, 21, and 24, as well as in attitude items 17, 18, and 20. However, female-taught classes had significantly higher scores in attitude item 19 (confidence in knowledge of climate change), as well as in knowledge items 22, 23, 25, and 26 and in behaviour item 27.

A further question was investigated: whether the relationship between performance and the gender of the student depends on the gender of the teacher. Table 3 shows male and female students' performance in the Toolkit activity by gender of the teacher.

The most salient finding relates to students' average scores for all items before and after participating in the Toolkit activity. When a class had a female teacher, girls achieved significantly higher overall scores than boys, both before and after the activity: 0.58 and 0.86 for girls compared with 0.49 and 0.74 for boys. When the class had a male teacher, boys achieved significantly higher overall scores than girls, both before and after the activity: 0.72 and 0.86 for boys compared with 0.34 and 0.72 for girls. Thus, there is a certain symmetry about the results, with the pattern for male teachers being the mirror image of the pattern for female teachers; the "after" average score for female students taught by female teachers (0.86) was identical to the "after" average score for male students taught by male teachers.

Independent samples *t*-tests showed that when mixed classes taught by female teachers carried out the Toolkit activity, female students achieved significantly higher "after" scores than male students, $t(26) = 2.63, p < .05$. Similarly, when mixed classes taught by male teachers carried out the Toolkit activity, male students achieved significantly higher "after" scores than female students, $t(26) = 3.52, p < .01$.

We can infer that in the context of the climate change activity carried out in Vanuatu, a teacher's gender does have an influence on student achievement, with both male and female teachers being more effective at teaching students of their own gender.

Differences in Performance by Gender Between Students in Urban and Rural Schools

A further question on performance by gender is whether there was a difference between students in urban and rural schools. Of the five schools in the sample, two are located in the capital city of Port Vila, with a total of 75 females and 44 males. The remainder, located in rural areas, have a total of 37 females and 53 males. In all, the sample of 209 comprises 119 students in urban schools and 90 students in rural schools.

Table 4 shows before and after scores for all females and males in urban schools with before and after scores for all females and males in rural schools. There is a clear pattern in the overall performance; in urban schools, female students outperformed male students in average scores both before the intervention (0.54 compared to 0.47) and afterwards (0.84 to 0.76), while in rural schools, males outperformed females beforehand (0.50 to 0.40) and afterwards (0.83 to 0.76).

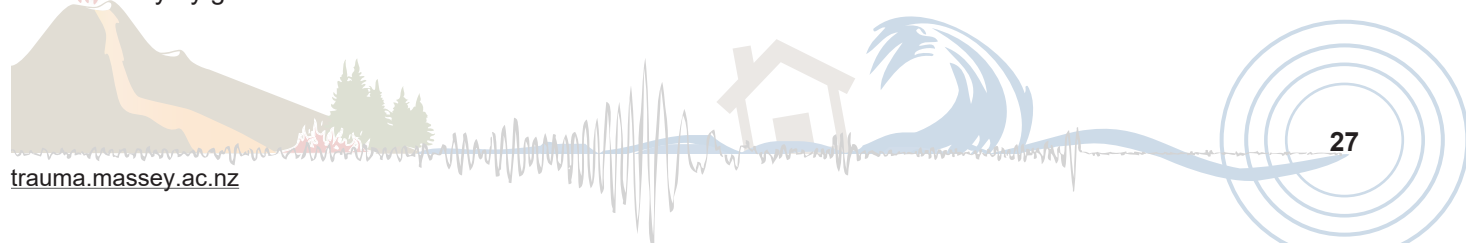


Table 2*Comparison of Average Scores for 209 Students Taught by Female and Male Teachers*

Statement / Question no.	Taught by female teachers (n = 127)			Taught by male teachers (n = 97)		
	Before Av. score	After Av. score	Change	Before Av. score	After Av. score	Change
1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.	0.91	0.80	-0.11	0.83	1.00	0.17
2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.	0.36	0.45	0.09	0.54	0.79	0.26
3. Our climate changes because of both natural and human factors	0.70	0.86	0.16	0.51	0.91	0.40
4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.	-0.17	-0.28	-0.10	-0.20	-0.18	0.01
5. Climate variability means that the climate of a place may change from year to year.	0.30	0.57	0.28	0.15	0.61	0.46
6. During an El Niño period, Vanuatu experiences droughts.	0.69	0.69	0.00	0.40	0.77	0.37
7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.	0.28	0.56	0.28	0.37	0.93	0.56
8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.	0.21	0.65	0.44	0.33	0.62	0.29
9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.	0.25	0.44	0.19	0.04	0.45	0.41
10. Carbon dioxide is absorbed by forests and the oceans	0.30	0.58	0.28	0.38	0.62	0.24
11. Coral reefs are being damaged by warmer temperatures and ocean acidification.	0.55	0.61	0.06	0.23	0.77	0.54
12. Vanuatu's greatest climatic dangers are cyclones and droughts.	0.89	0.89	0.00	0.62	0.91	0.29
13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.	-0.14	-0.34	-0.20	-0.15	-0.63	-0.49
14. Using compost is a sustainable form of gardening.	0.76	0.87	0.11	0.46	0.85	0.39
15. Catching fish with nets that have very small holes is a sustainable form of fishing.	0.06	0.26	0.20	-0.18	0.20	0.38
16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.	0.51	0.76	0.25	0.41	0.70	0.28
17. I am worried that climate change will bring great dangers to the world in the future. *	0.55	0.80	0.24	0.51	0.93	0.41
18. I want to join a student strike to show my concern about climate change.*	0.74	0.67	-0.07	0.63	0.88	0.24
19. I know a lot about the causes and impacts of climate change.*	0.21	0.60	0.39	0.07	0.39	0.32
20. I want to help my community prepare for climate change and disasters.*	0.82	0.86	0.04	0.63	1.00	0.37
21. What does "climate change" mean?	0.46	0.76	0.29	0.38	0.87	0.49
22. What is the enhanced greenhouse effect?	0.60	0.91	0.31	0.44	0.80	0.37
23. What are 3 human activities putting extra GHGs into the atmosphere?	1.71	2.39	0.69	0.74	1.60	0.85
24. What causes ocean acidification?	0.35	0.69	0.35	0.43	0.78	0.35
25. Why is forestry important for mitigating climate change?	0.74	1.25	0.51	0.61	1.06	0.45

	Before Av. score	After Av. score	Change	Before Av. score	After Av. score	Change
26. What are some of the main impacts of climate change in Vanuatu?	0.67	1.52	0.85	0.55	1.10	0.55
27. What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters?	1.30	3.02	1.72	1.38	2.71	1.33
Knowledge	0.50	0.72	0.22	0.36	0.71	0.35
Attitudes	0.58	0.73	0.15	0.46	0.80	0.34
Behaviour	1.30	3.02	1.72	1.38	2.71	1.33
All	0.54	0.81	0.27	0.41	0.79	0.38

Note. * = Attitude questions; Item 27 = Behaviour question; all other items = Knowledge

Table 3

Comparison of Average Scores for Female and Male Students Taught by Female Teachers with Those Taught by Male Teachers

	Female teachers				Male teachers			
	Female students (n = 74)		Male students (n = 53)		Female students (n = 38)		Male students (n = 44)	
	Before	After	Before	After	Before	After	Before	After
Knowledge (1-16)	0.4	0.55	0.42	0.49	0.25	0.53	0.34	0.63
Attitudes (17-20)	0.67	0.73	0.46	0.73	0.42	0.74	0.5	0.85
Knowledge (21-26)	0.86	1.35	0.61	1.12	0.4	0.92	0.63	1.14
Behaviour (27)	1.36	3.35	1.21	2.57	1.13	2.56	1.6	2.84
All	0.58	0.86	0.49	0.74	0.34	0.72	0.48	0.86

Table 4

Average Scores for Female and Male Students in Urban and Rural Schools Before and After Participating in the Toolkit Activity

	Urban schools				Rural schools			
	Female students (n = 75)		Male students (n = 44)		Female students (n = 37)		Male students (n = 53)	
	Before	After	Before	After	Before	After	Before	After
Knowledge (1-16)	0.37	0.53	0.39	0.49	0.29	0.55	0.37	0.61
Attitudes (17-20)	0.64	0.68	0.45	0.64	0.48	0.84	0.50	0.90
Knowledge (21-26)	0.78	1.32	0.57	1.19	0.54	0.97	0.66	1.07
Behaviour (27)	1.43	3.41	1.11	2.89	0.89	2.41	1.68	2.53
All	0.54	0.84	0.47	0.76	0.40	0.76	0.50	0.83

A similar pattern is observable in the results for knowledge (A/D items and short answer questions), attitudes, and behaviour. In all categories, and both before and after the intervention, girls outperformed boys in urban schools, but boys outperformed girls in rural schools.

Independent samples *t*-tests showed that there is a significant difference between average “after” scores of female and male students in urban schools, $t(26) = 2.23$, $p < .05$. However, the difference between average “after” scores of female and male students in rural schools was not statistically significant.

Seeking Reasons for the Difference in Performance by Gender Between Urban and Rural Schools

To explain why females scored significantly higher scores than males when the CC Toolkit activity was conducted in urban schools, but not in rural schools, two reasons can be suggested. Firstly, on purely statistical grounds, females constituted a higher proportion (63%) of the students in the sample from urban schools, and of the three teachers involved, two were female. In contrast, males made up 59% of the students in the sample from rural schools, and of the five teachers involved, three were male. As we have already seen from the results

on the influence of teacher gender, girls perform better when their teacher is female, and this might be one reason for the results shown in Table 4. Secondly, and of greater relevance, there is the likelihood that boys in urban schools are more distracted from their studies than girls. To explore this, email interviews were conducted with three secondary school principals, seven secondary school teachers, and one undergraduate at the University of the South Pacific.

Of these respondents, seven were male and four female, and all were ni-Vanuatu with experience of living in either Port Vila or Luganville. The question asked of them was: "From your experience of living in towns, would you say that boys at Year 9/10 level are more distracted or more disengaged from their studies than girls? What are the things going on in town that distract boys more than girls?"

Of the 11 respondents, nine had no doubt that boys are more easily distracted than girls:

I think boys are much more distracted than girls. From my experience, girls generally are very committed in studying. Boys on the other hand are easily distracted by peers, which influences them to be involved in activities such as drinking, smoking tobacco and cigarettes at a very young age, going to nightclubs and some even creating gangs to steal in the shops and plenty more.

Teacher of English, Whitewood High School

Boys seemed to be more distracted than girls due to the following influences:

1. Peer pressure at school or from dropouts within their neighbourhoods. They would rather hang around with them doing drugs and drinking. This has sometimes led to gang fighting in schools as well as high absenteeism from schools. They come to school but spend the time hanging around school without attending classes.
2. Smoking marijuana and drinking alcohol.
3. Distractive entertainments such as clubbing, partying and ceremonial activities (weddings, deaths, etc.)
4. Parental neglect, which leads to a high amount of distractions especially with boys. This neglect could be due to parental divorce, single parenting or busy working parents who do not spend enough time

with the children, so creating a sense of loss in the children that makes them neglect school activities or lose interest in their education, with school becoming a pastime activity while transitioning to adulthood.

5. Distraction caused by social media (including Facebook) and the internet, which leads to a lot of unnecessary entertainment that can mislead most boys to think that life is all about entertainment. So when lessons are not entertaining the boys become bored and disengaged.

6. The school curriculum does not provide a wholesome education that caters for students' needs and abilities and does not provide the learning they need. The curriculum is mostly theoretical and not practical enough, so only a few will be interested.

Teacher of Social Science, Nagavika College

Boys are more distracted than girls. The biggest distraction today is the online video game PUBG (Players Unknown Battle Ground). Boys want to do little chores to earn themselves money for purchasing phone data. They play this game at home till late hours and even in school during breaks, lunch time or free periods. In addition to this we have TikTok, Facebook Madlpz and FruityLoops, Remix/Music Software, which are more suited to boys than girls.

Teacher of Social Science, Acacia Secondary School

Two respondents, however, had differing views:

No, girls are more easily distracted. Their distractions are drugs, boy-girl relationships, social media (especially Facebook), and physical appearance.

Deputy Principal, Namariu Junior Secondary School

Girls are more easily distracted because nowadays they usually have boy friends at an earlier age than in the past due to social platforms.

Teacher of Social Science, Banyan School

Table 5 summarizes respondents' perceptions of the main factors causing boys to be more easily distracted than girls from their studies at school in urban contexts.

In short, these interviews lend substantial support to the statistical finding that in urban schools, girls performed significantly better than boys in the Toolkit activity because of the many factors causing boys to become disengaged from formal education.

Table 5

Teachers' Perceptions of Factors Causing Boys to Have Greater Disengagement from School than Girls in Urban Areas of Vanuatu

Factor	Number of respondents reporting this factor (out of 11)
Taking alcoholic drinks	8
Smoking cigarettes/tobacco	6
Drinking kava	4
Taking marijuana/illegal drugs	4
Peer pressure, especially from those who have left school	4
Involvement in gangs	3
Going to night clubs	3
Entertainment provided on social media, internet, movies, etc.	3
Stealing	2
Boys have more freedom than girls	2
Mobile phones	2
Lack of parental care	2
Higher participation in sport than girls	1
Boy-girl relationships	1
Influence of swearing	1
Inadequate school curriculum	1
Online video game PUBG	1

Discussion

This paper aims to examine the link between gender and climate change education through the use of the Climate Change Toolkit in Years 9-12. A sample of 112 female and 97 male students from eight classes in five secondary schools in Vanuatu participated in this discovery learning classroom activity. Measurement of changes in their knowledge, attitudes, and behaviour was obtained through the use of the same diagnostic questionnaire completed by respondents before and after undertaking the activity – a questionnaire comprising 20 agree/disagree items and seven open-ended short answer questions. Analysis of results shows that overall scores obtained for all items in the questionnaire were almost identical both before and after the intervention, with 0.49 for both genders beforehand and 0.81 for females and 0.80 for males afterwards. The same parity of performance was demonstrated by the magnitude of improvement between before and after scores. For knowledge, boys slightly outperformed girls in the agree/disagree items, while girls outperformed boys in the short answer questions. For attitudes, the Toolkit activity improved male scores more than female ones, while it was the reverse for behaviour, with females showing a higher degree of improvement.

The activity appears to have improved girls' performance in the short answer questions to a greater extent than

for boys. Six of these questions test knowledge about the nature and impacts of climate change, the enhanced greenhouse effect, ocean acidification, the role of trees in mitigating climate change, and anthropogenic causes of climate change – physical processes that boys in other situations are seen as grasping more readily than girls, as found by Kira and Komba (2015) and Ballew et al. (2018). If girls improved their scores more than boys in these six questions because of participating in the Toolkit activity, one inference is that the Toolkit itself was more helpful for girls in raising their scientific knowledge than it was for boys. Another inference is that it was equally effective for both sexes but that teenage girls had greater commitment to study (Hadjar et al, 2014; Jackman & Morrain-Webb, 2019; Van Houtte, 2004) and perhaps higher capacity to express their newly acquired knowledge in written form than did teenage boys. Similarly, the seventh short answer question, on intended behaviour, may have captured females' greater motivation to look for ways to care for their families and communities in the face of climate change and more intense disasters, in keeping with the findings of Zelezny et al. (2000) – a motivation stimulated by Toolkit pictures 9 and 10 (on hydro-meteorological disasters) and 11-16 (on strategies for adaptation).

Regarding the effect of teacher gender on students' performance, average scores for all items before the

intervention were significantly higher in those classes taught by female teachers than in those taught by males, but when students carried out the activity, those taught by male teachers then attained similar scores to those taught by females. One reason could be that male teachers were more effective in implementing the Toolkit activity. Another might be that the students taught by female teachers already knew more of the material under study. A third possibility is that the small sample size led to a distortion of results, and that with a much larger group of students and teachers there would have been no significant change in overall performance between students taught by male and female teachers.

When disaggregating results by domain of learning, however, a different pattern emerged. For both knowledge and attitudes, students in mixed classes taught by male teachers showed a greater improvement in scores than those taught by females. For behaviour, that pattern was reversed, with female-taught classes showing a more striking improvement in intention to adopt a greater variety of adaptation measures. This could be because female teachers have a heightened risk perception of environmental dangers (Bord & O'Connor, 1997; Xiao & McCright, 2012) and because of their natural tendency to stress the importance of caring for family and community (Zelezny et al., 2000).

For individual items, a similar reasoning could apply to the finding that female-taught classes achieved significantly higher scores than those taught by males for showing confidence in knowledge of climate change (item 19), in knowledge of the enhanced greenhouse effect and its causes (items 22 and 23), in explaining the role of trees in absorbing atmospheric carbon dioxide (item 25), in stating a range of climate change impacts in Vanuatu (item 26), and describing actions that bring greater resilience (item 27). Such results relate to McCright's (2010) view that women are more likely than men to have greater knowledge about climate change and to worry about its effects. As female teachers, therefore, this knowledge and worry may be communicated directly or indirectly in their classrooms, perhaps influencing students' performance in the Toolkit activity.

A significant finding has been that in terms of performance, the gender of the student and gender of the teacher are clearly linked, with both male and female teachers being more effective at teaching students of their own gender. The pattern for male teachers was the mirror image of that for female teachers: with male teachers, average scores for boys both before and after the intervention were higher than those for girls, while with female

teachers, girls had higher average scores than boys beforehand and afterwards. The findings support those of Muralidharan and Sheth (2016), who based their conclusions on 5 years of data from primary schools in the Indian state of Andhra Pradesh. However, they were looking at performance across all school subjects, and at levels corresponding to Years 1 to 6 in Vanuatu's educational system, while this study focuses on just one classroom activity carried out by an older cohort of mostly Year 9-10 students.

In the final aspect of this paper's investigations - differences in performance by gender between urban and rural schools - another clear pattern emerged, with females achieving statistically significant higher scores than males when the Toolkit activity was conducted in urban schools, while there was an apparent (but non-significant) trend of males outperforming females in rural schools. One reason for this pattern could be that the girls and female teachers accounted for the majority of the sample in urban schools, and results have already demonstrated that girls perform better when their teacher is female. On the other hand, boys and male teachers made up a higher proportion of the sample in rural schools. A second reason comes from my own observations through 35 years of teaching in Vanuatu that distractions in urban areas have a greater influence on boys than girls. I therefore decided to conduct email interviews with 10 teachers and one university student with experience of urban living in Vanuatu and triangulate their views with the quantitative data collected from students.

Of the 11 respondents, nine confirmed that in an urban environment, Year 9/10 boys are more easily distracted from their studies than girls. The most common factor involved was perceived as the consumption of alcoholic drinks, closely followed by smoking cigarettes and alcohol. Other important influences are the consumption of kava, marijuana, and other illegal drugs, usually associated with peer pressure from boys in the neighbourhood who have already dropped out of school. In some cases, membership in a gang becomes all-consuming and involves a boy in stealing and going to night-clubs. In general, respondents said that it is customary for teenage boys to be accorded greater freedom than girls, with the latter under stricter parental control and required to return home straight after school to assist with household duties and continue with their studies. The influence of social media was seen to affect both boys and girls in urban areas, but was of less significance in rural schools, where internet access may not be readily available.

These findings are in line with research in Caribbean schools by Jackman and Morrain-Webb (2019). They also support the findings of Hadjar et al. (2014) that boys are more alienated from school than girls, leading to “non-conformity and anti-social behaviour in the school environment” (p. 121) and lower educational achievement. This study drew on literature in a European context but is relevant to Vanuatu in that its two urban centres will offer a similar intensity of distractions to that found in a Western environment – which in turn will affect boys more than girls and perhaps explain girls’ superior performance in the Toolkit activity when undertaken in towns.

On a broader note, this study has implications for building the resilience of Vanuatu’s young people to climate change and disasters. The nation’s two key policy documents on resilience – the Vanuatu Climate Change and Disaster Risk Reduction Policy 2016-2030 (VCCDRRP) (Government of Vanuatu and SPC, 2015) and Vanuatu 2030: The National Sustainable Development Plan (Government of Vanuatu, 2016) – refer in general terms to the importance of educating students about resilience issues. Environmental goal # 3.4 of the VCCDRRP aims to “promote and ensure strengthened resilience and adaptive capacity to climate related, natural and man-made hazards” (p. 14). The VCCDRRP says that climate change and disaster risk reduction have been mainstreamed into the National Curriculum Statement. However, the implementation of revised curricula since 2010 has been very slow, and for junior secondary school students, the Toolkit was in 2022 one of the few resources that can promote their resilience. This paper has demonstrated that both girls and boys at junior secondary school level benefit from exposure to the Toolkit and associated activity, with average overall scores rising for both boys and girls, and a marked improvement in scores for behaviour intended to bolster adaptation, particularly in girls.

The impact on girls’ knowledge, attitudes, and behaviour, especially when their teacher is female, may perhaps contribute towards the VCCDRRP’s goal of greater empowerment for women in climate change issues:

Climate change and disaster risks impact greatly on the lives of women and their social and economic wellbeing. In disaster preparedness, as well as response and recovery, women make enormous contributions to the welfare of their communities. It is vital that women have full opportunities to participate in policy development, decision-making

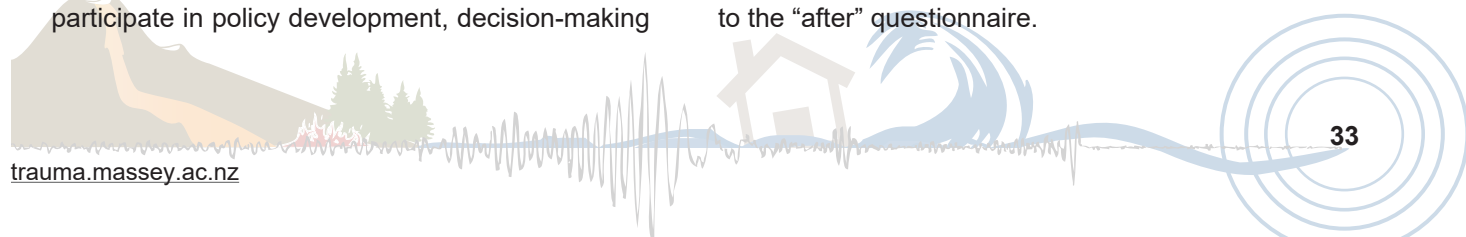
and implementation at all levels. Initiatives are under way to ensure that gender considerations are mainstreamed into all climate change and disaster risk reduction efforts, and efforts should be increased. Women have historically been excluded from participating on committees and other decision-making forums, and their full representation needs to be assured.

(Government of Vanuatu & SPC, 2015, Section 8.2, p. 26)

In the long-term, however, the usefulness of the CC Toolkit may decline as new syllabi are introduced at junior secondary level which reflect the UN’s Sustainable Development Goals as well as the VCCDRRP and Vanuatu’s National Sustainable Development Plan. In this way, students and teachers should have access to a richer range of pedagogy and materials that help to build “a stable, sustainable and prosperous Vanuatu” (Government of Vanuatu, 2016, p. 1) and ensure greater resilience to climate change for the country’s youth.

Limitations of the Research

This study on the effectiveness of the Climate Change Toolkit on male and female resilience to climate change and disasters has several limitations. Firstly, the sample included only 209 students from eight classes. The larger parent sample of 363 students and 19 classes from which information was collected to assess the influence of other variables, although representative of males and females and urban and rural schools, French- and English-medium classes, and a range of five year levels, could not be used for gender-specific data because this had not been planned at the outset of the research. Secondly, investigations were carried out during a period of stress and uncertainty for teachers and students because of the state of emergency arising from COVID-19 and, initially, Cyclone Harold. Schools were closed during the first few months of this period, and then when they did resume, teachers did not prioritize the task of undertaking this classroom intervention on climate change. Restrictions on international travel also imposed limitations on the researcher’s ability to interact with teachers and students involved. Thirdly, it is likely that the activity was not always undertaken in the prescribed manner, because of shortage of time. In particular, the learning that would occur through class discussion of the pictures once the activity had been completed may not in fact have taken place in several classrooms. This would impact upon the accuracy of students’ answers to the “after” questionnaire.



Conclusion

As part of my overall research into the effectiveness of formal learning about climate and disaster resilience in Vanuatu, I examined the situation at the junior secondary level of education, focusing on the usefulness of "Learning About Climate Change the Pacific Way" and its accompanying discovery learning activity. As the research unfolded, it became possible to make comparisons between the performance of girls and boys, and between girls and boys in rural and urban settings. Evidence from questionnaires completed by a sample of 112 girls and 97 boys from eight classes, drawn largely from Years 9 and 10, demonstrates that when average scores for knowledge, attitudes, and behaviours were aggregated, there was no significant difference in performance between males and females either before or after the intervention. On the other hand, when the gender of the teacher was considered, female students performed better overall when their teacher was female and male students achieved higher scores when their teacher was male. A closer examination of the situation in urban and rural schools confirmed this same trend: in urban schools, female students outperformed males both before and after the intervention, while in rural schools males outperformed females, although to a lesser extent and not statistically significantly.

This data was collected during the first 15 months of the COVID-19 pandemic, with the researcher having to collect information by remote means. This was a particular challenge when endeavouring to evaluate reasons for the differences observed between male and female performance, with email interviews having to replace face-to-face contact with teachers and students. Nevertheless, the emerging evidence suggests that the distractions of living in an urban area exert a greater influence on boys than on girls, with a consequent tendency for urban boys to become more disengaged from formal education, mainly because of the influence of alcohol, cigarettes, kava, marijuana, peer pressure from those who have already left school, greater personal freedom, and accessibility to the internet.

Teachers and students said that they appreciated the manner in which the cooperative and discovery learning strategies of this intervention engaged their students, dealt with real-life situations in a Pacific Island context and appeared to have an impact on students' knowledge, attitudes, and intended patterns of behaviour. Yet teachers made no comments on students' misunderstandings of mitigation and adaption nor of scientific processes involved in the enhanced

greenhouse effect or the absorption of carbon dioxide by forests and oceans.

A tentative conclusion to draw from the findings is that while the Toolkit had a similar effect on the overall performance of both genders, cultural factors in Vanuatu suggest that girls' resilience to climate change will be better when their teacher is female, and boys' resilience will improve when the teacher is male. Additionally, boys' interest in becoming resilient to climate change is more likely to occur when they are being educated in a rural environment that has fewer distractions than those of the city and can more readily observe the changes in the environment that are already resulting from disasters and the climate crisis. Although this study was specific to Vanuatu, some of the findings may have implications for other Pacific Island nations.

Finally, this study demonstrates the usefulness of the Climate Change Toolkit in contributing towards the broad aims of Vanuatu's two key policies on CCDRR and sustainable development. It has raised the competency of boys and especially of girls to gain more knowledge of the scientific aspects of climate change, have greater concern over its impacts, be more willing to help their community prepare for climate change and disasters, and to undertake actions that will forge greater adaptive capacity. In time, its usefulness may lessen as new syllabi and materials become available, but for the moment, it is a proven educational resource for Vanuatu.

Recommendations

It is recommended that further studies be done on how the performance of students in the Climate Change Toolkit activity may be affected by gender of student and gender of teacher, but that such studies involve a much larger cohort of students and teachers and include a more systematic element of initial teacher training. This has cost implications for bringing teachers together in a central location in each of Vanuatu's provinces, and also for the provision of picture sets and updated teacher guides.

There may also be a further, wider issue that needs to be addressed: how to effect changes in Vanuatu culture whereby a person's gender does not define the way in which he/she acquires or imparts knowledge, skills, attitudes, or desired behaviours, so removing the effect of gender bias on learning.

Disclosure statement

The author was the Vanuatu contributor to the set of pictures Learning about Climate Change the Pacific Way during its development in 2012-2014. He has 35 years of teaching experience in educational institutions in Vanuatu.

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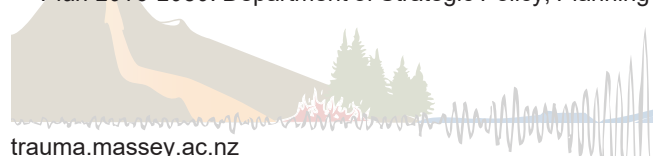
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Appendix

Form for tabulating numerical scores for responses

Name of school and class:

Date:

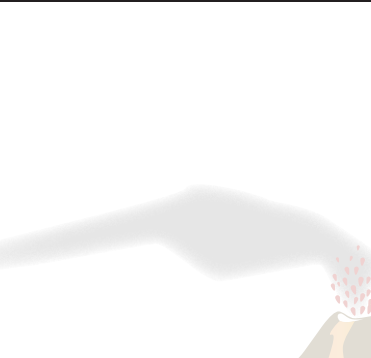
Number of students participating (n):

Name of teacher:

Questions asked before or after use of Toolkit pictures? (Encircle): **BEFORE** AFTER

	Statement	+ 1 (correct) - 1 (incorrect)	My opinion				
			Agree	Disagree	Don't know	Total score(T)	Av score (T ÷ n)
1.	1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.						
2.	2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.						
3.	3. Our climate changes because of both natural and human factors						
4.	4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.						
5.	5. Climate variability means that the climate of a place may change from year to year.						
6.	6. During an El Niño period, Vanuatu experiences droughts.						
7.	7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.						
8.	8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.						
9.	9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.						
10.	10. Carbon dioxide is absorbed by forests and the oceans						
11.	11. Coral reefs are being damaged by warmer temperatures and ocean acidification.						
12.	12. Vanuatu's greatest climatic dangers are cyclones and droughts.						
13.	13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.						
14.	14. Using compost is a sustainable form of gardening.						
15.	15. Catching fish with nets that have very small holes is a sustainable form of fishing.						
16.	16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.						
17.	17. I am worried that climate change will bring great dangers to the world in the future.						
18.	18. I want to join a student strike to show my concern about climate change.						
19.	19. I know a lot about the causes and impacts of climate change.						
20.	20. I want to help my community prepare for climate change and disasters.						

	Question	Answers
21.	What does "climate change" mean? 5: Accurate explanation of the concept and processes involved (e.g. significant <u>long-term change</u> in global climate or average weather patterns) 3: Satisfactory level of understanding of the processes involved (e.g. change in climate or change in average weather conditions or patterns) 1: Limited level of understanding (e.g. change over a period of time; it is getting hotter; there are longer droughts) 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	(Student answers are recorded here, then a score of 0-5 given to each one and all scores totalled for the question)

22.	What is the enhanced greenhouse effect? 5: Accurate explanation of the concept and processes involved (e.g. an <u>increase in greenhouse gases</u> in the atmosphere due to <u>human activities</u>, which leads to <u>greater atmospheric heating</u>) 3: Satisfactory level of understanding of the processes involved (e.g. an increase in dangerous gases in the atmosphere; increase in gases that absorb/trap heat radiated back from the earth) 1: Limited level of understanding (e.g. gases that cause global warming; gases absorbing the heat radiated back from earth; effects caused by GHG such as CO₂, methane, etc; effects such as melting ice, sea level rise, destruction of plants and animals, warmer temperatures) 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
23.	What are three human activities that are putting extra greenhouse gases into the atmosphere, so causing the enhanced greenhouse effect? Valid activities are: • Burning of fossil fuels in vehicles/ships/ aircraft • Burning of fossil fuels in industry/ factories • Deforestation • Cattle rearing • Rice cultivation. • Garbage/rubbish tips, burning of rubbish/plastic 5: Accurate: Three valid activities identified 3: Satisfactory: Two valid activities identified 1: Limited: One valid activity identified 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
24.	What causes ocean acidification? 5: Accurate explanation of the process involved (e.g. absorption of more and more CO₂ by the oceans; oceans acting as carbon sinks; absorption of CO₂ by oceans causes weak carbonic acid; increase of CO₂ in the oceans, which lowers pH value) 3: Satisfactory level of understanding of the processes involved (e.g. carbon dioxide is absorbed by the oceans) 1: Limited level of understanding (e.g. carbon dioxide; carbonic acid; methane, ozone, CO₂; warmer temperatures; pollution by oil/fuel/chemicals/sewage) 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
25.	Why is forestry (planting and caring for trees) important for mitigating climate change? 5: Accurate explanation of the concepts and processes involved (e.g. trees absorb carbon dioxide from the atmosphere, so reducing atmospheric CO₂ content, so slowing/ lessening global warming and climate change.) 3: Satisfactory level of understanding of the processes involved (e.g. trees absorb carbon dioxide [and give out oxygen]; trees reduce the amount of CO₂ in the atmosphere) 1: Limited level of understanding (e.g. adaptation measures: reduce soil erosion; provide habitat for living organisms; provide	

	<i>shade; provide materials for house construction and medicines; protect the environment; support livelihoods, etc.)</i> 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
26.	What are some of the main impacts of climate change in Vanuatu? <i>Valid impacts are:</i> • Sea level rise • Coastal erosion • Damage to coral reefs/aquatic life • Ocean acidification • Longer periods of drought • Loss of crops due to drought or higher temperatures • Higher temperatures / hot temperatures • More extreme weather events, e.g. stronger cyclones, more floods • Change in fruiting/flowering seasons of trees and plants • Other, e.g. distress, increase in diseases such as malaria and dengue 5: Accurate: At least 3 valid impacts 3: Satisfactory: Two valid impacts 1: Limited: One valid impact 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
27.	What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters? List 3. <i>Answer should give adaptation strategies appropriate at family or community level for preparing for CC / more intense disasters, e.g. plant more trees/reafforestation, build stronger houses, get larger water tanks, re-locate buildings away from the shoreline or a river, plant mangroves, use composting and mulching, use solar energy/panels, plant and eat more aelan kakae, plant crop varieties that are resilient to CC and disasters, encourage more gardening, establish community conservation areas/ marine conservation areas, use environmentally-friendly fishing methods (e.g. canoes), conserve/save water, conduct CC awareness, encourage greater unity in the community. Mitigation measures can be mentioned, e.g. use of renewable resources, use public transport rather than private vehicles, recycling, avoid burning rubbish/plastic, etc. Preparations for disasters can also be given, e.g. food preservation, gathering food and water, cyclone-proof housing, etc.</i> 5: Accurate: Three valid actions identified 3: Satisfactory: Two valid actions identified 1: Limited: One valid action identified 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	

Example of a summary data form for one className of school and class: Sandalwood JSS Yrs 9/10 Science Date: 22nd March 2021

Number of students participating (n): 21 Name of teacher:

Questions asked before or after use of Toolkit pictures? BEFORE AFTER

	Statement + 1 (correct)  - 1 (incorrect) 	Agree	Disagree	Don't know	Total score (T)	Av score (T ÷ n)
1.	1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.	14	2	5	12	0.57
2.	2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.	16	2	3	14	0.67
3.	3. Our climate changes because of both natural and human factors	9	5	7	4	0.19
4.	4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.	3	10	8	7	0.33
5.	5. Climate variability means that the climate of a place may change from year to year.	8	5	8	3	0.14
6.	6. During an El Niño period, Vanuatu experiences droughts.	11	3	7	8	0.38
7.	7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.	7	6	8	1	0.05
8.	8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.	9	4	8	5	0.24
9.	9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.	4	5	12	-1	-0.05
10.	10. Carbon dioxide is absorbed by forests and the oceans	9	5	7	4	0.19
11.	11. Coral reefs are being damaged by warmer temperatures and ocean acidification.	6	2	13	4	0.19
12.	12. Vanuatu's greatest climatic dangers are cyclones and droughts.	5	7	9	-2	-0.10
13.	13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.	4	4	13	0	0.00
14.	14. Using compost is a sustainable form of gardening.	8	6	7	2	0.10
15.	15. Catching fish with nets that have very small holes is a sustainable form of fishing.	4	10	7	6	0.29
16.	16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.	8	5	8	3	0.14

Knowledge

17.	17. I am worried that climate change will bring great dangers to the world in the future.	8	6	7	2	0.10
18.	18. I want to join a student strike to show my concern about climate change.	7	6	8	7	0.33
19.	19. I know a lot about the causes and impacts of climate change.	3	10	8	-7	-0.33
20.	20. I want to help my community prepare for climate change and disasters.	6	7	8	-1	-0.05

Attitudes

21.	What does "climate change" mean?	2	0.10
22.	What is the enhanced greenhouse effect?	6	0.29
23.	What are 3 human activities putting extra GHGs into the atmosphere?	16	0.76
24.	What causes ocean acidification?	5	0.24
25.	Why is forestry important for mitigating climate change?	19	0.90
26.	What are some of the main impacts of climate change in Vanuatu?	14	0.67

Knowledge

27.	What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters? List 3.	27	1.29
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Behaviour

Experiences of children impacted by the 2019/2020 bushfires in South Australia: A qualitative study

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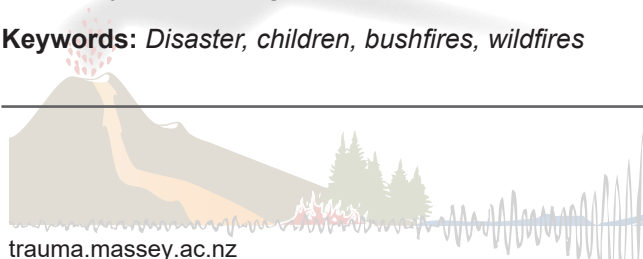
Abstract

Little is known about children's subjective experiences of bushfires. This study explores child experiences of South Australian bushfires that occurred during 2019/2020. Semi-structured interviews explored the experiences of children who lived through the fires in the Adelaide Hills (n = 7) and Kangaroo Island (n = 2) and were subsequently referred to a tertiary mental health service. Responses were thematically analysed from a psychological perspective. Five themes were generated: (1) planning, (2) separation, (3) fear of fire's destruction, (4) sense of a close call, and (5) the fire left its mark. Children reported similar bushfire experiences, and recalled the event as chaotic and frightening. Many experienced brief separations from caregivers without means to communicate and conveyed a deep concern for others' safety including family and pets. Clinical care practitioners should be aware of the specific fears and worries children experience during disasters, and particularly, whether children experienced separation from a caregiver. Furthermore, the presence and effect of pre-existing, co-occurring negative life events and secondary stresses ought to be considered.

Keywords: Disaster, children, bushfires, wildfires

In 2019, prolonged drought, above-average temperatures, and a series of dry lightning storms eventuated in a string of devastating climate-induced bushfires in Australia. Between November 2019 and February 2020, South Australia was severely impacted with almost 280,000 hectares burnt, 211,500 of which were on Kangaroo Island (Government of South Australia, 2020). In South Australia, the bushfires, commonly referred to as wildfires internationally, resulted in the deaths of three people, the destruction of 196 homes, the deaths of nearly 68,000 livestock, and substantial impacts on native habitats and wildlife loss (e.g., 50,000 koalas on Kangaroo Island; Government of South Australia, 2020). Furthermore, across South Australia, approximately 8,000 people were displaced throughout the bushfire event (Du Parc & Yasukawa, 2020). Whilst exact duration of displacement is unknown, it would be expected to range between a few hours to days and beyond especially given that bushfire recovery efforts have shown that house rebuilding can take between 1- 4 years (Australian Red Cross, 2020).

Children, young people, and adults may have similar disaster experiences, such as disruption and displacement, perceived threat, and harm or loss of life and belongings. Several methodological approaches have been used to understand and represent children's experiences and their post-disaster needs. These approaches include comparative observations by carers of infants residing in London in war nurseries with those living in typical circumstances (Freud & Burlingham, 1944), and adult (e.g., parent/teacher) completed questionnaires about disaster-related mental health impacts on children (see Spencer & Thompson's, 2024 scoping review). Children's participation in disaster research includes their completion of self-report mental health screeners and standardised questionnaires (e.g., Geng et al., 2019; Yelland et al., 2010). Furthermore, activities such as focus groups, walking tours, and photo-elicitation techniques have been used to garner children's views about their experiences and resilience building skills: for example, their knowledge and contributory roles to disaster responses and policies (e.g., Gibbs, Waters, et al., 2013; Walker, 2016). The CUIDAR (Cultures of Disaster Resilience among children



and young people) European Union-funded projects are examples of such endeavours (Mort et al., 2020).

Children may be disproportionately affected by their disaster experiences such as changes in their usual supports and social networks and ongoing disruptions that affect their safety and stability (Danese et al., 2020; Gibbs et al., 2015; Norris et al. 2002; Raphael & Burns, 2014). Short- and long-term effects include higher risk than adults of developing post-traumatic stress disorder (PTSD; Le Roux & Cobham, 2022; McDermott & Palmer, 2002; To et al., 2021; Yelland et al., 2010), depression (Brown et al., 2019), and longer-term psychological morbidity (McFarlane et al., 1987; McFarlane & Van Hoof, 2009). Given that surge demand for psychosocial services can outstrip organisational capacity, a stepped-care model that aligns needs with treatment intensity (i.e., universal, targeted, and specialist interventions) has been recommended (McDermott & Cobham, 2014; Phoenix Australia, 2020).

Provided through the Women's and Children's Health Network in South Australia, the Child and Adolescent Mental Health Service (CAMHS) is a state-wide tertiary mental health service which provides care to young people with severe and/or complex mental health needs. In response to the bushfires, CAMHS received funding from the Government of South Australia, via the Office of the Chief Psychiatrist, to create a multi-disciplinary Bushfire Response Team which provided services to bushfire-affected young people. Given the effect of other negative life events and co-morbidities on post-disaster trajectories and recovery (La Greca et al., 2013; Lai et al., 2013), referrals were accepted from key community agencies, including schools, and from parents/carers of children continuing to experience distress related to the fires and/or the extensive media coverage. The current study aimed to explore children's subjective experiences of the 2019/20 South Australian bushfires by analysing data from questionnaires and clinical interviews undertaken at least 6 months post-bushfires by the CAMHS Bushfire Response Team.

Methods

Participants and Setting

A total of 140 families from fire-affected regions were serviced by the CAMHS Bushfire Response Team, of which 13 families consented to participate in a research study. Nine participants are included in the current study. Four participants were excluded as questionnaires were answered by caregivers, rather than the child.

Participants were aged 5 to 14 years ($M = 10.33$, $SD = 2.44$), seven participants resided in the Adelaide Hills, and two on Kangaroo Island. Four participants were girls and five were boys. Given location and population size and to preserve participants' anonymity, further identifying details are not provided.

Data Collection

This study was approved by the Women's and Children's Health Network's Research Ethics Committee (2020/HRE01641). Data collection took place between August 2020 and September 2022. The data used in the current study involved caregivers and children completing questionnaires (e.g., Child PTSD Symptom Scale for DSM-5 (CPSS-V); Foa et al., 2018). Children wrote their responses to the CPSS-V and were supported by the interviewer who wrote their dictated responses in some cases if preferred by the child. The other data was from a semi-structured interview conducted by the treating clinician with the child, in the presence of a caregiver. This was recorded by note taking by the interviewer who was in all cases the treating clinician. Additional questionnaires that yield quantitative data were administered but are not included here as they are beyond the scope of the current study (see Appendix).

The CPSS-V requires the child to nominate a "scary or upsetting thing" that bothers them most when they think about it and then rate frequency and severity of reactions. Children were asked to self-nominate bushfire and non-bushfire related experiences. The semi-structured interview explored child and family experiences during the bushfire. The interview was devised by Dr Anne Gannoni (third author) based on available literature and in consultation with CAMHS staff and external experts and took approximately 40 minutes to complete. Most interviews were conducted face-to-face with one completed via telehealth. Questions included details about evacuation, damage to belongings, worries and fears during and after the event, involvement during fires, knowledge of and comparisons with peers' experiences, and the experience of returning home (see Appendix for semi-structured interview content).

The clinical interview was to be administered on intake, with the CPSS-V to be administered on intake, every 3 months during service engagement, and on discharge. However, there were discrepancies amongst participants in questionnaire and interview completion. Seven of the nine participants completed the clinical interview. Regarding the missing data - a clinical interview was completed only with parents for one child, and an

interview was not conducted with the remaining child. Eight children completed the CPSS-V. Completion of measures can be found in Table 1.

Table 1

Participant Completion Status of the Clinical Interview and CPSS-V

Participant	Clinical interview	CPSS-V
1	Not completed	Completed three (intake, 3 months, and discharge)
2	Completed	Completed two (intake and discharge)
3	Completed	Completed one (intake)
4	Completed	Completed two (intake and discharge)
5	Completed	Completed one (intake)
6	Completed	Not completed
7	Completed	Completed two (intake and discharge)
8	Not completed	Completed one (intake)
9	Completed	Completed one (intake)

Data Analysis

Reflexive thematic analysis (TA; Braun & Clarke, 2022) was selected to explore child experiences of the fires. Given its bottom-up analytical approach, reflexive TA allows for an inductively-oriented analysis, focused on finding patterned semantic and latent meanings. The analysis was conducted by the first author. The process involved familiarisation with and thematical analysis of the data, which included the child's self-nominated bothersome bushfire and non-bushfire experiences (CPSS-V; Foa et al., 2018) and responses to the semi-structured interview. Using NVivo12 software for coding, the data set was worked through systematically four times, each time in a different order to avoid an unevenly coded data set (Braun & Clarke, 2022). Initial candidate themes were generated, reviewed, and refined. The final themes that were derived were based upon their relevance to the research aim and discussed with two supervising researchers with background knowledge in the research project and thematic analyses.

Results

Five themes were generated through the analysis process: (1) planning, (2) separation, (3) fear of fire's destruction, (4) sense of a close call, and (5) the fire left its mark (Refer to Table 2 for themes and sub-themes developed through thematic analysis).

Table 2

Themes and Subthemes Developed Through Thematic Analysis

Theme	Subthemes
Planning	Evacuation plans as reactive Evacuation plans as proactive
Separation	Communication during evacuation
Fear of fire's destruction	Fear of losing belongings Fear of losing the house Fear for others' safety
Sense of a close call	
Fire left its mark	Lasting environmental impact Lasting psychological impact

Planning

Children were asked about where they were when they heard about the fires and what subsequent actions were taken. Children provided details about whether they had an evacuation plan and if they adhered to that plan. One participant described not feeling worried during the evacuation as they followed their family's bushfire plan:

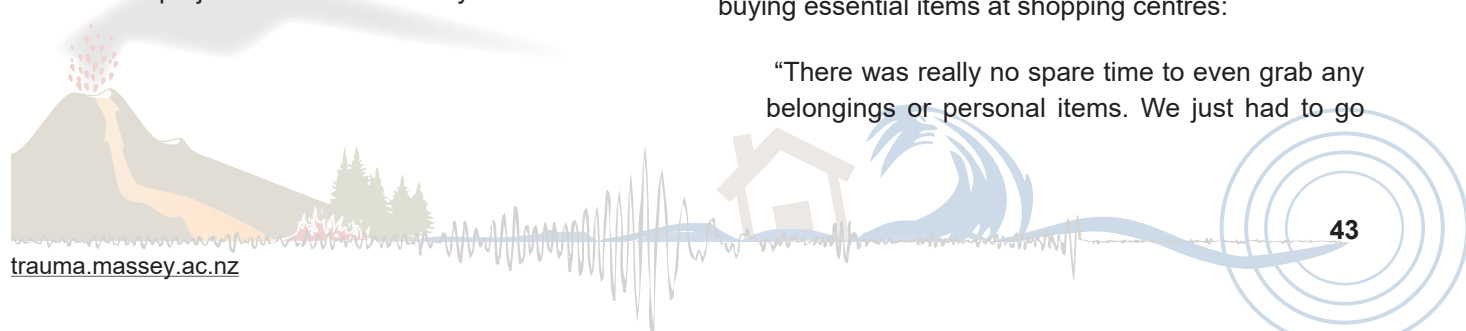
"Not worried. I knew we were at Grandma's and fires didn't go that way and we were safe there and that's where we go if we need to evacuate."

Overwhelmingly, however, accounts from the children describe that there was not enough time to enact fire plans and commonly experienced a rushed and reactive evacuation. For example, an interviewer noted the following about a child's experience:

"Family had planned to leave, though left it too late to leave and took refuge on the sports oval where people were using wire cutters to open the gates".

Children reported that original plans had to be modified due to the rapidly evolving threat. This is implied by the children's accounts of having to leave their homes quickly, often without belongings, and subsequently buying essential items at shopping centres:

"There was really no spare time to even grab any belongings or personal items. We just had to go



quickly.” Additionally, “[we] had to go to [shopping centre] for toothbrushes and phone chargers as we couldn’t go home to get belongings”.

These experiences create a sense that the fire danger evolved faster than anticipated or that evacuation was unexpected. This is again reinforced by children recounting that they were away from their homes when fire warnings were issued, and thus unable to follow original bushfire plans:

“Family has plan for when at home, but no plan for if out and have to evacuate”.

Additionally: “[I was] watching Star Wars at cinema. Mum got all these messages when we got out. Went go grandma’s [...] no belongings.”

These accounts demonstrate that children were aware of their family’s bushfire plans, and aware if they were not followed. Children conveyed that many of their families were caught off guard by the fire. This is further supported by most children recalling their parents had left for work on the morning of the fires. This led to most children being separated from family during the evacuation process, with separation discussed next as another main theme.

Separation

During evacuation, most children experienced separation from a family member at some point. Across participant age groups, periods of separation were commonly described as eliciting “worried” and “scared” emotions, with children often not knowing where their family were and whether they were safe. For example, one child stated:

“Can’t remember if Dad was at work. Worried about where he was. Better when [Dad] returned because everyone there to hug me because I was crying and sad.”

While they were separated, children described attempting to keep in contact with family members, with communication emerging as a subtheme. Participants described having to listen to the “scanner”¹ and “radio”, as they had “no power”, “no internet”, and “no service”. This made communicating with separated family members difficult and sometimes impossible during evacuation. One child stated:

“Mum at home till left for work [...] lost communication with mum on freeway”.

Once evacuated, many of the children’s parents left to help in different capacities, such as assisting neighbours, fighting the fires, or collecting supplies. Children conveyed immense fear for their parents’ safety, for example:

“I left and mum tried to put fire out. I was scared I would never see my mum again”.

Overall, separation occurred for varying reasons and at different points of the evacuation process. However, regardless of these differences, the experience of separation was commonly described as causing children distress and eliciting concern for the safety of the people from whom they were separated. Indeed, many children spoke about fearing for others’ safety during the fires, demonstrating a keen awareness of the fire’s destructive potential. Fear of the fire’s destruction was subsequently generated as a main theme.

Fear of the Fire’s Destruction

Children commonly described fear of the fire’s destructive capabilities, including the risk it imposed on the safety of their family members. During the semi-structured interview, children were asked if they were worried about their family members being hurt. All of them indicated that they were and listed the people they were worried about. Children listed their mum, dad, siblings, grandparents, and pets. Children responded that they were: “worried about dad”, “scared I would never see my mum again”, and “mum getting stuck” and “dad getting more hurt”. In addition, four of the seven children who completed interviews (which took place from 10 months to 2 years post disaster), rated their worry about others’ welfare at the time of the fires in the very high range (i.e., 8-10, with 10 labelled as “Heaps worried”). Also taken at interview, their ratings reflecting their current worry about others’ welfare were placed in the mid-range (i.e., 3-7), suggesting ongoing concern despite the time-lapse. In contrast, when asked if they were worried for themselves at the time (e.g., getting hurt), several children responded “no” or did not respond, with one participant explaining that their “mum and dad had it under control”. On the other hand, two children indicated that they were very worried for their personal safety (i.e., ratings 8-10, with 10 labelled as “Heaps worried”). Children’s descriptions of the psychological effects of their experiences are included in the theme, “The Fire left its Mark”.

Multiple children described concerns for their pets they had to leave behind. One child wrote she feared her pet would “wander into the fire”. Another consistently

¹ A scanner is a radio receiver that can automatically tune, or scan two or more frequencies permitting connection to services, (e.g., Country Fire Service) and businesses.

referenced his concerns for the family dog: “worried and scared for Buster²” and still “I worry about Buster a lot, that he’s going to die”. Another child conveyed his concern for the local koalas, describing how he cared for them after the fires:

“[I] helped save koalas in my cart [...] I saved lots of them, gave water, but they were very sick. Some went to the wildlife park.”

These accounts indicate that the children had an acute awareness of family members and animals being in a life-threatening situation. In addition, most children conveyed fears of losing their homes, with children commonly saying that “the house burning down” was one of the scariest thoughts they had during the fires.

In addition to their homes, children referenced fear of losing their personal belongings. For example, one child described the fires as:

“Really scary, worried about my stuff, didn’t want it to burn down”.

It is evident the children knew of the fires’ potential to harm people, animals, and their homes, and this was a common fear shared amongst participants. The majority of participants’ homes were unscathed by the fire, but many commented on the proximity of the fire to their properties, conveying they felt it was a “close call” and a worse outcome had been anticipated. This was subsequently identified as a main theme.

Sense of a Close Call

A key theme generated was a shared sense amongst most participants that they narrowly escaped the fire’s destruction. This theme derived from children’s descriptions of their feelings of relief and gratitude when returning home implies they had expected or imagined losing their houses and belongings. For example, when asked about their return home, one child said:

“Relieved to see house to believe it wasn’t gone [...] Almost did, fire came to grass”.

Similarly, an interviewer noted the following about another child’s experience:

“Very emotional, saw devastation, very upset. Filmed roadside on the way home. Grateful to still have a house.”

One child gave details of the fire’s proximity to their home and compared their experience to that of their neighbours:

“House opposite- gardens and veranda burnt. House diagonally- opposite burnt, 200 meters away- and all of the trees. If the CFS (Country Fire Service) were not there, would have lost house.”

This account demonstrates a child contextualising their experience in relation to their neighbours and using it as an example of what they themselves narrowly avoided. Additionally, many children described evidence of the fire being near their homes, such as ash in their backyards and swimming pools, bins melted on the side of the road, and views of “blackness” from their homes. These descriptions, coupled with reported emotions of relief and gratitude, imply many children felt it was a close call regarding the survival of their homes and belongings as well as their parents or pets.

The Fire Left its Mark

After the immediate threat of the bushfires had passed and families returned home, children described how the fires left a lasting mark, both environmentally and psychologically. Many participants commented on the changed landscape near their homes with one interviewer noting the following about a child’s description of the fire damage:

“Regrowing but can still tell where the fires were, trees still more sparse”.

Indeed, many children described being surrounded by visual reminders of the bushfire’s impact, for example, explaining their area lost “80% of trees”, or the environment is now “black” and “burnt” with “nothing around”.

In addition to the physical marks left by the fires, children described persisting psychological impacts. One child whose house “totally burnt down” despite parental efforts to save it, expressed that since the fire, they become worried when separated from their mother:

“I worry when I’m not with my mum. I worry that if I feel sick or have a tummy ache or feel nausea coming on I can’t trust anyone else to care for me.”

Two children described recurring nightmares about the fires: “I get nightmares about the fires, and I can hear the birds crying from burning.” Another child, who had been co-sleeping with parents since the bushfires, stated: “[I]

2 Name changed for confidentiality.

Dream fire coming over the back, need to get out [as] fire gets closer.”

Naturally, the fires were a topic of discussion at school, for example, one child stated:

“Back at school - people talking about it A LOT. Talking about what they did”.

Further to the child’s account about koala rescues, two other children described how they helped others, one through volunteering with family at the donation centre sorting clothes and the other:

“[Sibling] and me have a few friends affected by the fire so we took extra handouts, and gave to our friends”.

Most children said they knew of community members who had suffered burns or other physical and mental health problems or had died since the fires through non-bushfire related events. To ascertain the effect of the co-occurring COVID-19 pandemic on children, a 2-item questionnaire to be administered at the interview had been included. Children’s ratings about the extent of their worry in the past month were placed in the lower range [i.e., 0 = not at all to 5 = (somewhat) worried] with the “things [they] most worried about” being “getting sick”, “not seeing family”, “not seeing friends”, and “not being able to do my usual things”. Moreover, some children’s self-nominated current bothersome non-bushfire related worries/experiences included: continued bullying at school for over 12 months, the unexpected death of a family friend, and thoughts of significant harm to and death of parents and self. Notably, some children in our study reported symptoms congruent with PTSD criteria such as nightmares. This suggests the possibility of persistent and potentially exaggerated negative beliefs about the world being more dangerous. Overall, children’s accounts demonstrate the bushfire’s persisting physical reminders and psychological effects, with co-occurring stressors and negative events (including the COVID-19 pandemic), maintaining hypervigilance, and a sense of threat, harm, and mortality to self and others.

Discussion

This study aimed to explore the experiences of South Australian children who were impacted by the 2019/2020 bushfires and referred to a tertiary mental health service. Previous research has predominantly focused on the development and presence of specific mental health disorders in young people, with their

subjective experiences often unexplored or reported upon (Self-Brown et al., 2017). Yet recognising children’s perspectives, especially the meaning and the effects of disasters, may promote their sense of value and belonging to community and importantly inform organisational service delivery to meet their needs.

This study extends existing knowledge through analysis of children’s accounts of their experiences during the fires and uniquely represents child voices from a tertiary mental health service context. Children indicated that overall, their experiences of the bushfires were chaotic and frightening, whether it was while families prepared to leave, subsequent evacuation, separation, or upon their return home. Experiences of separation were reported as common and distressing. Additionally, concern and awareness of “others” was a salient thread that wove together aspects of the children’s reported experiences. “Others” include parents, siblings, pets, local animals, and by extension, the wider living environment. The reactions of some children were further amplified by their non-bushfire experiences which included pre-existing and co-occurring secondary stressors such as the COVID-19 pandemic.

Loss of Communication During Evacuation

Brief separation was found to be a common and stressful occurrence, with children highlighting struggles to stay in contact with family members due to damaged infrastructure. Richardson et al. (2016) posited that society’s collective reliance on mobile devices may amplify the feelings of disconnection and uncertainty when they falter during such events, particularly for children who are sometimes referred to as “digital natives” (Bittman et al., 2011). Indeed, uncertainty is known as a contributing factor to anxiety (Gu et al., 2020) and, therefore, it stands to reason that reducing aspects of uncertainty may reduce some distress experienced by children in these circumstances. For example, several children mentioned their parents using other forms of technology to stay informed of the developing situation like the radio or the “scanner”. Given children’s status as “digital natives”, they may be well positioned to learn and instruct others about alternative technologies. Indeed, in their study investigating children’s experience of a bushfire education program, Gibbs et al. (2018) quoted one child who had since taught their mother to use radios, demonstrating their ability to give appropriate advice to parents. Although uncertainty and separation appear inevitable in the context of bushfires, children within our study point to ways this may be reduced. Previous studies also offer support for involving children

in these processes (e.g., Pfefferbaum et al., 2018, gives examples of children's participation in activities across preparedness, response, and recovery stages).

Brief separation

Although established as common and stressful, research about experiencing separation during disasters has largely explored the impact of longer-term separation caused by displacement (Hansel et al., 2013; Lau et al., 2010; McFarlane et al., 1987; Richardson et al., 2016; Usami et al., 2012). Yet, due to the unpredictable and rapidly developing nature of the bushfire, seeking safety from attachment figures was at times thwarted for the children in this study. Additionally, they shared stories of separation and concern for their pets, which are also regarded as attachment figures (Groenewoud et al., 2023). However, despite reunification, children reported ongoing anxieties about their pets and family. Thus, the potential of brief separation to lead to changes in attachment style is important to consider, as such disruptions have been linked to poorer behavioural, academic, and mental health outcomes (Schröder et al., 2019). As such, Bryant (2023) has recommended that models of PTSD incorporate attachment theory and social processes into biological, learning, and cognitive processes, given their important contribution to PTSD aetiology, maintenance, and outcome.

Social Comparison

Aside from their own bushfire experiences, children provided examples of the experiences of "others" (e.g., neighbours, school mates, or local animals), some of whom had been injured, lost their homes, or passed away. Drawing upon others' experiences and measuring it against their own may demonstrate a process of social comparison (Festinger, 1954). Downward comparisons made with others who are in the same or less fortunate situations generally occur when there is a threat to self-esteem and can serve self-protective and enhancement purposes (Wills, 1981). For adults within a post-disaster context, the role of social comparison as a cognitive coping strategy contributed, as highlighted by Brady et al. (2021), to the establishment of explicit and implicit hierarchies of affectedness. In turn, these had the potential to influence personal and community consequences such as contrasting validations of experience, external agencies' responses, and the availability of resources including entitlements (Brady et al., 2021). These authors also noted that for some survivors, these processes served as a helpful sense-making tool, whereas for others, it would be a source of

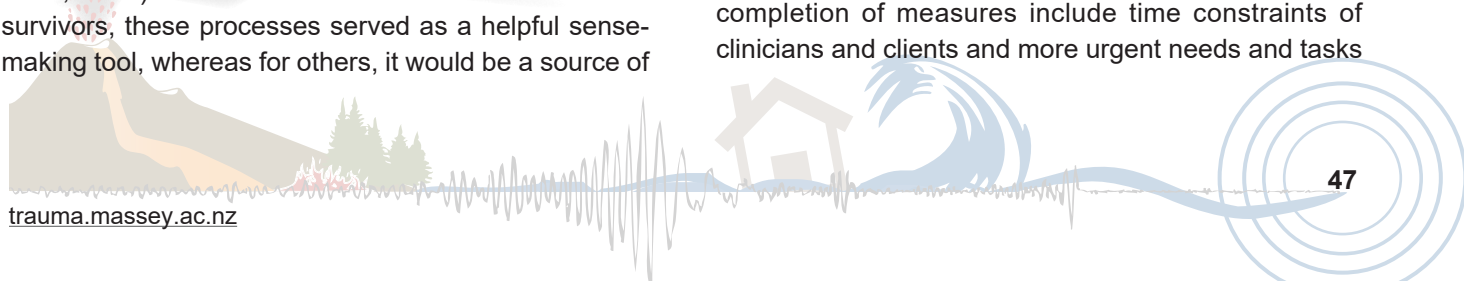
secondary stress whereby they felt judged regarding how badly they were affected compared to others. In our study, children often referenced others who they deemed were "worse off" than themselves, demonstrating downwards social comparison. In contrast (i.e., reflective of upward social comparison), for one child who lost their house, a sense of anger and injustice is implied in their description of their affectedness relative to less impacted friends ("my house burnt down where my friends only had (to) evacuate"). Like other research (e.g., Mort et al., 2018) our theme, "sense of a close call", may have captured part of this social comparison process, whereby for most children, their comparative perspectives demonstrate feelings of relief for their position post-disaster, alongside more complex realisations relating to threats, mortality, and safety.

Changes to the Environment

Coinciding with their reports of others' experiences, children often referenced changes to their home environment and places around them, including damage to neighbouring properties and their surroundings, using descriptors like "burnt" and "black". Their subjective accounts demonstrate that they indeed care for their local environments and animals, and damage or injury to them elicits distress. This may suggest vulnerability to experiencing solastalgia, a concept describing distress due to the perception and/or degradation of inhabited environments and the loss of comfort the environment once provided (Block et al., 2019; Hinds & Sparks, 2008; Leviston et al., 2023). Whilst not evident in this study, we acknowledge the broader scope of place connections (e.g., Scannell et al., 2016; Summers et al., 2019) and which for First Nations peoples relates inherently to spiritual and cultural attachments and practices (Breth-Petersen et al., 2023; Williamson et al., 2020).

Limitations and Strengths

Within the current study, several limitations that reflect challenges faced when conducting research in post-disaster settings were identified. These include a low participation rate and the method and completion of measures. Irrespective of methodological approaches (i.e., quantitative and qualitative), low participant numbers and recruitment challenges have been reported in many studies (e.g., Cobham & McDermott, 2025; Delicado et al., 2017; Spencer & Thompson, 2024). Several factors that likely affected the method and completion of measures include time constraints of clinicians and clients and more urgent needs and tasks



taking precedence for families during disaster recovery periods (Knack et al., 2006). However, despite these challenges, the current study attempted to make best use of responses available, ensuring the experiences were honoured as rich and informative sources of data. This can be taken as a relative strength of the study, whereby despite challenges posed by research in a post-disaster context, child voices were placed at the forefront.

Additionally, participants within the current study were receiving care from a tertiary mental health service and may therefore be representative of children who were more psychologically affected by the fires. As the intention of the study was to explore experiences within this cohort, it is an important point to consider when comparing current findings with other study populations. Including children's self-nominated non-bushfire worries and child-perceived impact of the COVID-19 pandemic further strengthened the study as previous research has shown that the experiences for adults (Cowlshaw et al., 2024) and adolescents (Beames et al., 2023) of bushfires and pandemics as compound disasters are disruptive of recovery processes.

Implications and Conclusion

This study heeds recommendations to adopt a child-centred approach made by researchers (Gibbs, MacDougall et al., 2013; Harms et al., 2021; Peek, 2008), government bodies (Australian Government National Emergency Management Agency, 2020), and non-government organisations (Royal Far West and UNICEF Australia, 2021). The study captures the complex cumulative experiences of some children who lived through the 2019/2020 South Australian bushfires and were referred to a tertiary mental health service. Children described separation and disrupted communication to be particularly distressing. This highlights potential practical and clinical implications: a need to mitigate and plan for separation and communication, how separation may be experienced by the child as a distressing psychological event, and what can be done retrospectively to repair this.

Turning now then to specialist level interventions, the Trauma-Focused Cognitive Behavioural Therapies (e.g., CBT and Eye Movement Desensitisation and Reprocessing) have the strongest evidence base (McDermott & Cobham, 2014; Phoenix Australia, 2020), but with limited interventions and evidence available for younger children and those with disabilities (Gibbs et al., 2021). However, Bryant's (2023) recommendation

to include attachment theory and social processes into PTSD models underscores the need to provide relational/attachment-based therapies. An example and included in the CAMHS Bushfire Response Team's intervention framework is Parallel Parent Child Narrative (PPCN), stage 1 of the 4-stage process of Parallel Parent and Child Therapy (P-PACT) developed by Heather Chambers. Still requiring formal evaluation, PPCN's key approach is working with parent/child dyads to undertake a form of therapeutic storytelling. The focus is on the relationship, on revealing mistakes of meaning, countering hurt, shame, and blame, and proving good intentions for parent and child. It has been used in New Zealand and Australia since the early 2000s across private practice and governmental services (including CAMHS) and found to be an acceptable and beneficial intervention by families and therapists (Amos, 2024; Furber et al., 2013). Furthermore, given that children engaged in social comparison after the fires, relational/attachment therapies may provide opportunities to support conversations about community hierarchies of affectedness.

The data analysed for the current study was drawn from a data set which included parent reports of the bushfire. Continuing with the theme of relational understandings, we recommend a comparison between parent and child reports as these may provide useful insights into how events are perceived from differing perspectives.

Considering children are particularly vulnerable to the threats posed by climate-induced disasters and the knowledge that these events are predicted to increase (United Nations Environmental Programme, 2022), it is pertinent to continue including children in research, especially when they are service recipients, and to look for ways to include them as co-researchers (Gibbs, Mutch, et al., 2013).

Acknowledgement

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Authorship Statement

Stella Lycos: Conceptualisation, data curation, formal analysis, methodology, visualisation, writing- original draft, writing- review and editing.

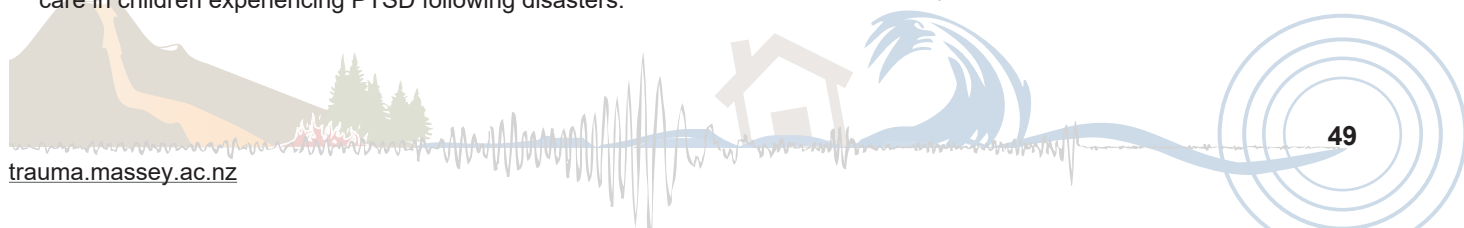
Rachel Roberts: Conceptualisation, project administration, resources, supervision, writing-review and editing.

Anne Gannoni: Conceptualisation, data curation, investigation, project administration, resources, supervision, writing-review and editing.

Susan Hamilton: Data curation, writing- review and editing.

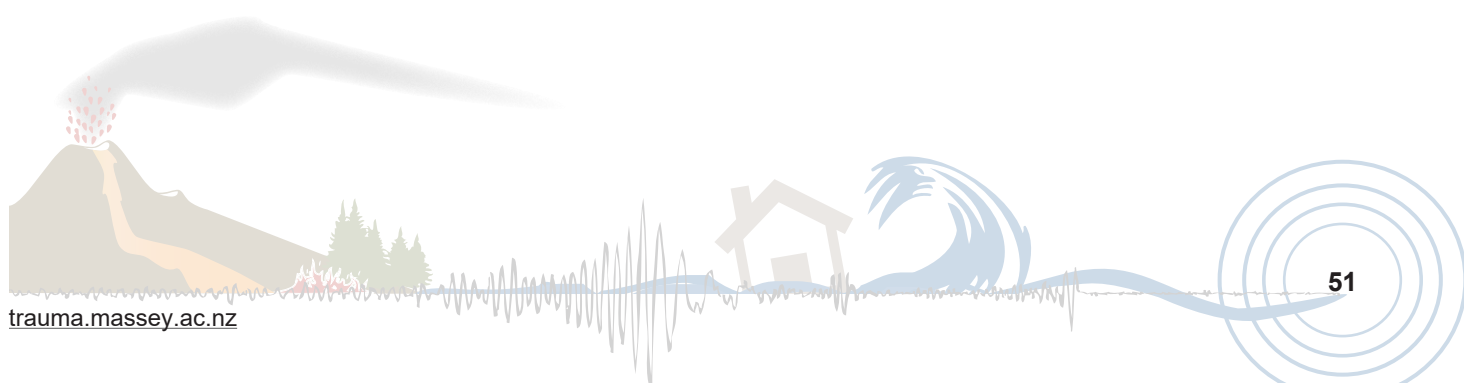
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Appendix

Protocol Questionnaires

- Initial consultation and allocated clinician appointments

Revised Child Impact of Events Scale (CRIES-8)

The Children's Revised Impact of Events Scale (CRIES-8) is a brief child-friendly measure designed to screen children at risk for posttraumatic stress disorder (PTSD; Perrin et al., 2005). It is a worldwide measure used following large and small-scale disasters, has good reliability, a stable factor structure and has been administered to children as young as 5 years old.

Child PTSD Symptom Scale (CPSS-V): Child and Parent Versions

The CPSS-V is a 20 item self-report measure of child PTSD for young people aged 8-18 years of age (Foa et al., 2018). A further seven items focus on daily function, (e.g., relationships with friends, schoolwork) to yield a total impairment score. The CPSS-V is reported to have excellent internal consistency and good test-retest reliability.

PTSD Checklist for DSM-5 (PCL-5)

The PCL-5 is a 20-item parent self-report measure that assesses the presence and severity of PTSD symptoms in adults in the past month (Weathers et al., 2013). Items correspond with DSM-5 criteria for PTSD and the measure takes about 5-10 minutes to complete and has sound psychometric properties.

Additional questions: Semi-structured interview

Questions to garner children's experiences of the fires were devised for the study. Although written in a semi-structured format, the questions are asked in a child-friendly conversational style and parent/caregiver could be present.

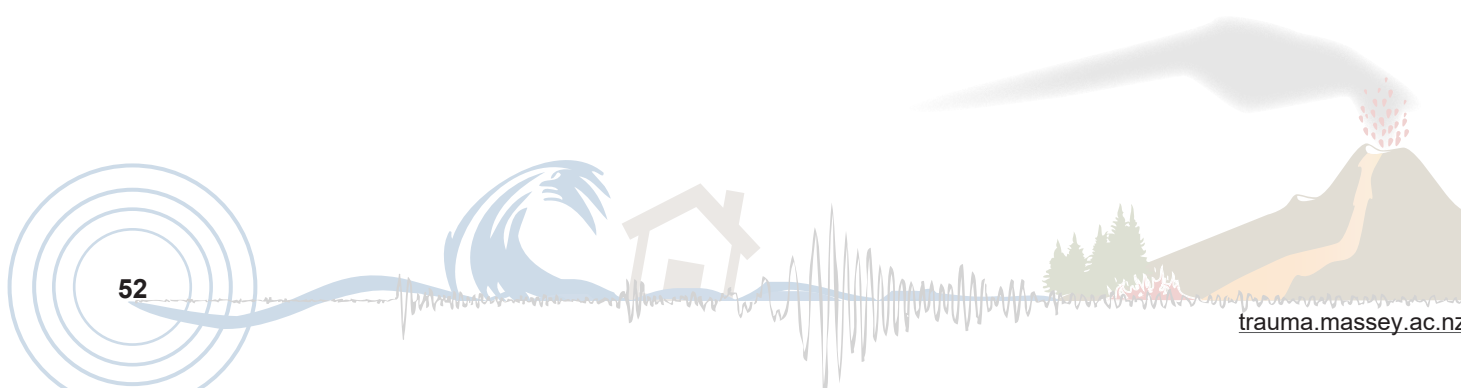
Two brief questionnaires were devised to be administered early in the service contact:

- 1) Covid 19, i.e., a worry rating and the things most worried about and
 - 2) Pre-Bushfire functioning: i.e. use of, and recommended need for psychosocial services pre-bushfire.
- Repeat administration at 3 months until discharge related to service delivery requirements.

CPSS-V (Child and Parent version), PCL-5, and semi-structured interview (reduced items depending on treatment phase and status).

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Additional questions

These questions cover several areas: evacuation, damage to belongings, worries and fears during and after the event, involvement during fires, knowledge of, and comparisons with peers' experiences, and the experience of returning home. The wording and sequence of questions and topics may be altered for relevancy and to promote child friendly conversation, and normal flow. (Some examples of follow up topics/questions are provided in brackets).

"I would like to talk with you about what happened for you and your family when the bushfires happened at Christmas time."

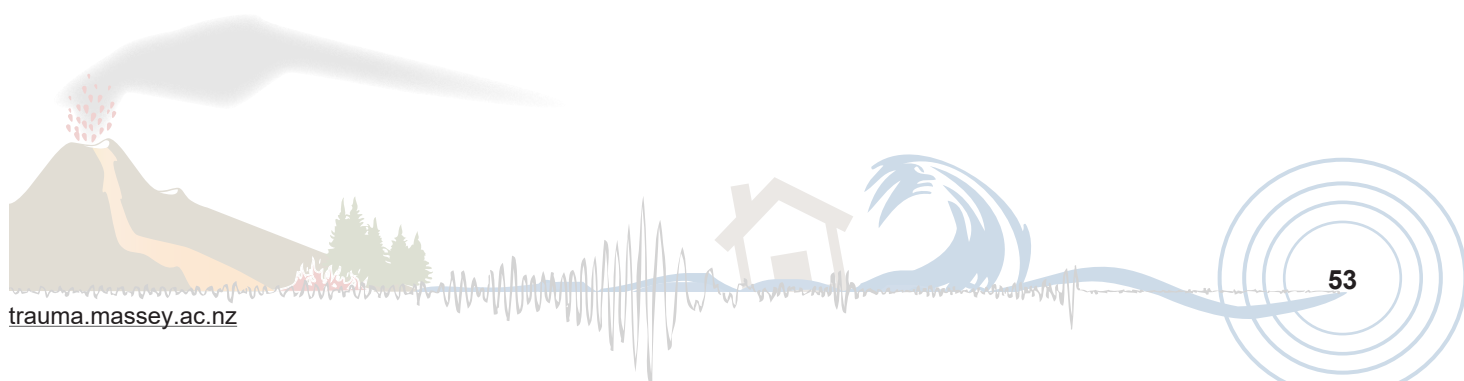
- Were you and your family at home when the bushfires happened or were you somewhere else?
- Using a map, identify location of home and proximity to fires. Record any other details.
- Did you stay at home, or leave when the bushfires were happening?
- At home, were you trapped and unable to leave, separated from parents/caregivers?

(What was that like for you, other members of the family? (e.g., who was at home, what did you/they do; If separated, who were you with, duration, hearing about family members whereabouts, welfare; what were your experiences/ reactions at the time, other family members reactions).

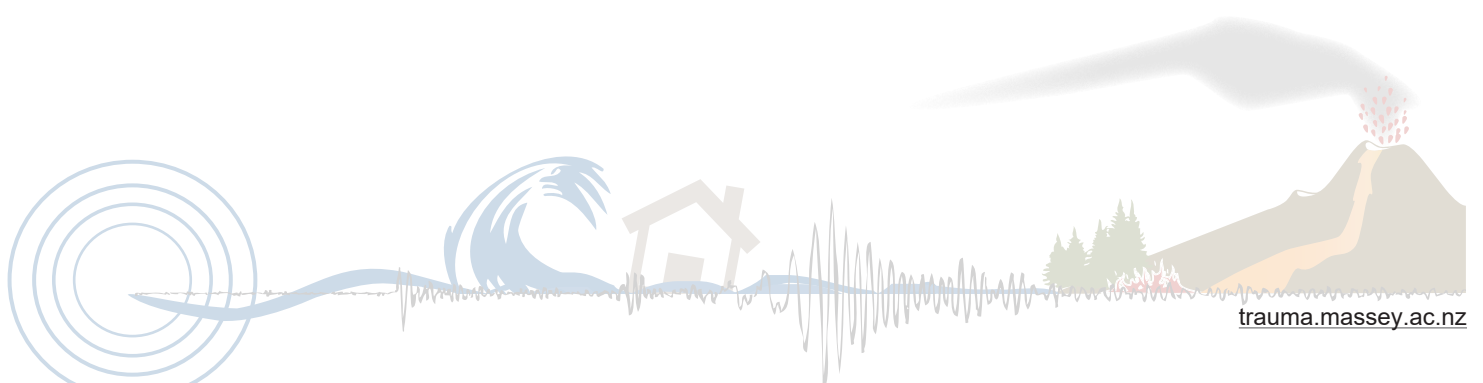
- If left, was that planned or unplanned? Please tell me what happened. (e.g., what did you/family members have to do?)
- If evacuated, location, duration of evacuation, with whom?
- Was any of your property damaged? (Describe damage/ loss and ask about child's belongings).
- Do you have any animals, pets/stock? what happened to the animals? What did you see?
- If damage and/or stock loss, what is happening now with your property/stock/land?
- Was any member of your family involved in helping with the fires? Were you involved?
- Did anyone you know get hurt in the bushfires? Tell me what happened.
- Were you worried that other members of your family would get hurt? Who? How worried were you about them? How worried are you now about them?
- Were you worried that you would get hurt? How worried were you? 0 (not at all) -10 heaps worried)
- Did you have the same or different experiences to your friends and classmates? In what way was it the same/different?

Follow up questions

- To what extent has life returned to normal for you? Rate 0 (not at all)-10 (back to normal)?
- Please tell me what has and hasn't got back to normal for you, your family.



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